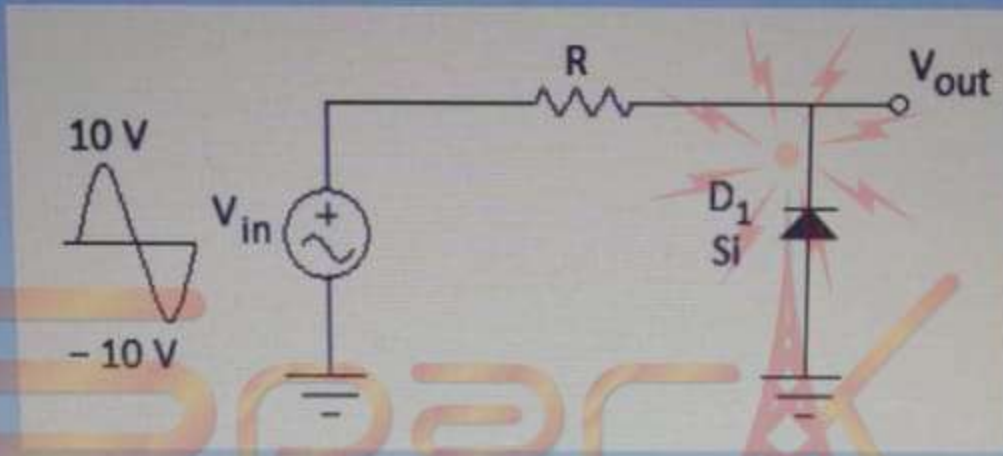
19

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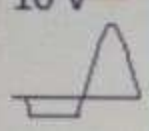
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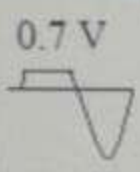
uestion

جهد المخرج للدائرة المبينة ادناه:

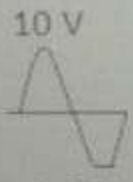


Select one:

a. 

b. 

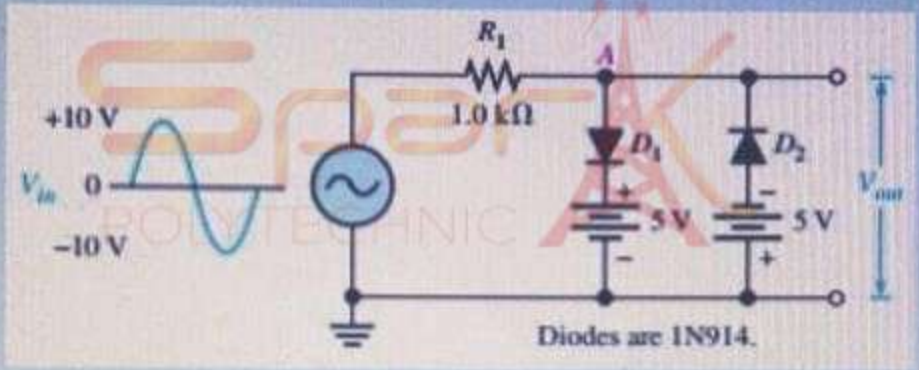
c. 

d. 



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For the clipper circuit below, the output will be: (in order)



Select one:

- a. 5.7,-5.7
- b. -5.7,-5.7
- c. -5.7,5.7
- d. 5.7,5.7

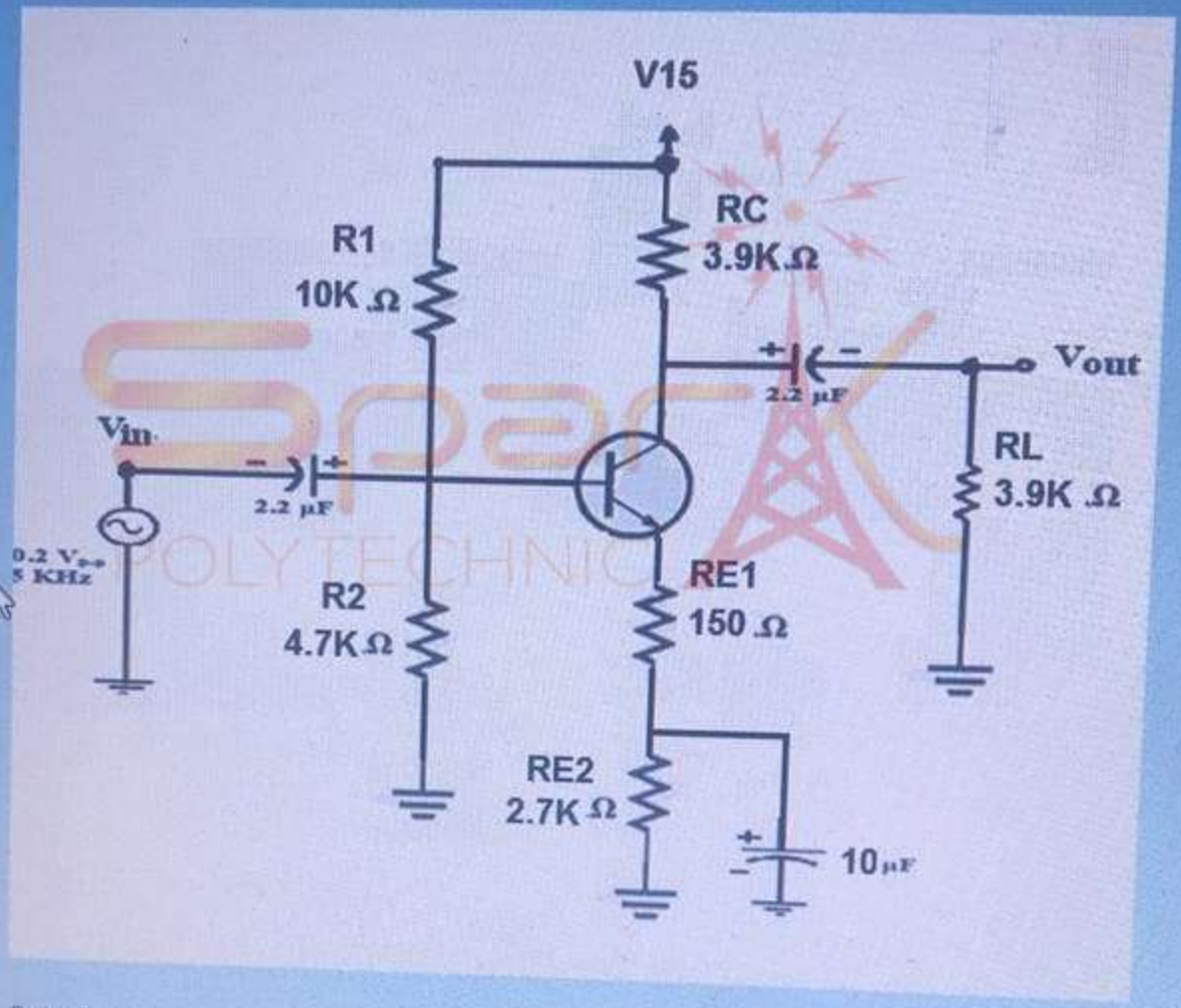
Zener diodes are also known as

Select one:

- a. Forward bias diode
- b. Breakdown diode
- c. Voltage regulators
- d. None of the mentioned

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For the circuit shown we use the Bypass capacitor in order to



Select one:

- a. increase the voltage gain
- b. no effect of the bypass capacitor

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LECTRONIC DEVICES AND CIRCUI

9

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خواص الانتقال للترانزستور

JFET

هو العلاقة بين

Select one:

- a. I_G, V_{DS}
- b. I_G, V_{GD}
- c. V_{DG}, I_D
- d. I_D, V_{GS}



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0112- ELECTRONIC
RCUITS LAB-3

عند فحص الديود باستخدام جهاز AVOmeter ظهرت قيمة 0 v على شاشة جهاز القياس.



فان diode يكون

Select one:

- a. في حالة فتح
- b. صالح للاستخدام
- c. لا يمكن تحديد وضعه
- d. في حالة قصر

B

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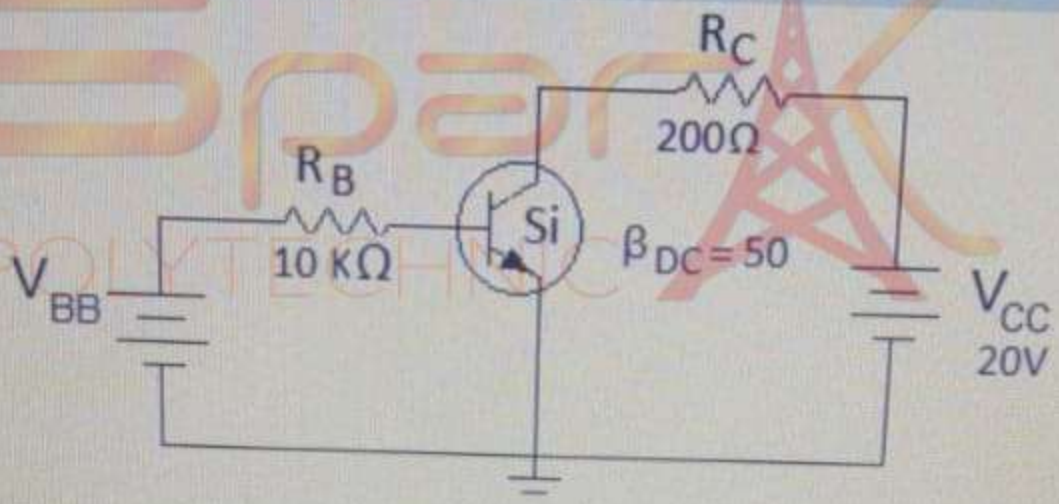
RCUITS LAB-3

إذا علمت أن جهد المصدر

$V_{BB}=0\ V$

فان قيمة تيار الباعث تساوي

I_E



Select one:

- a. 67 mA
- b. 50 mA
- c. 0 A
- d. 4.6 mA

C

Next page

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0112- ELECTRONIC
RCUITS LAB-3

عند استخدام مضخم باعث مشترك زاوية فرق الطور بين جهد الدخل و جهد الخرج تساوي

Select one:

- a. 90 درجة
- b. 0 درجة
- c. 360 درجة
- d. 180 درجة

D

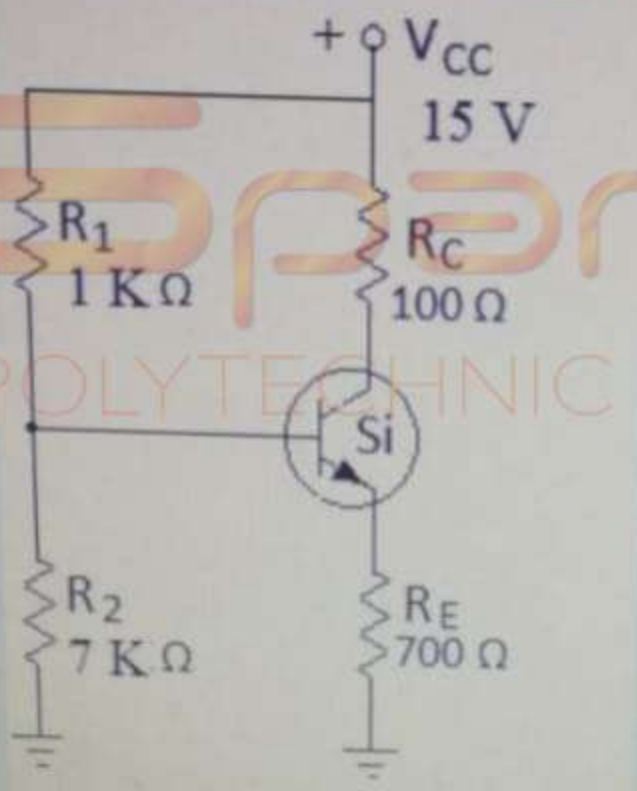
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RCUITS LAB-3

لدائره الترانزستور المبينة أدناه, فأن فرق الجهد بين القاعده و الباعث يساوي

V_{BE}



Select one:

- a. 18 V
- b. 0.7 V
- c. 6 V
- d. 10 V

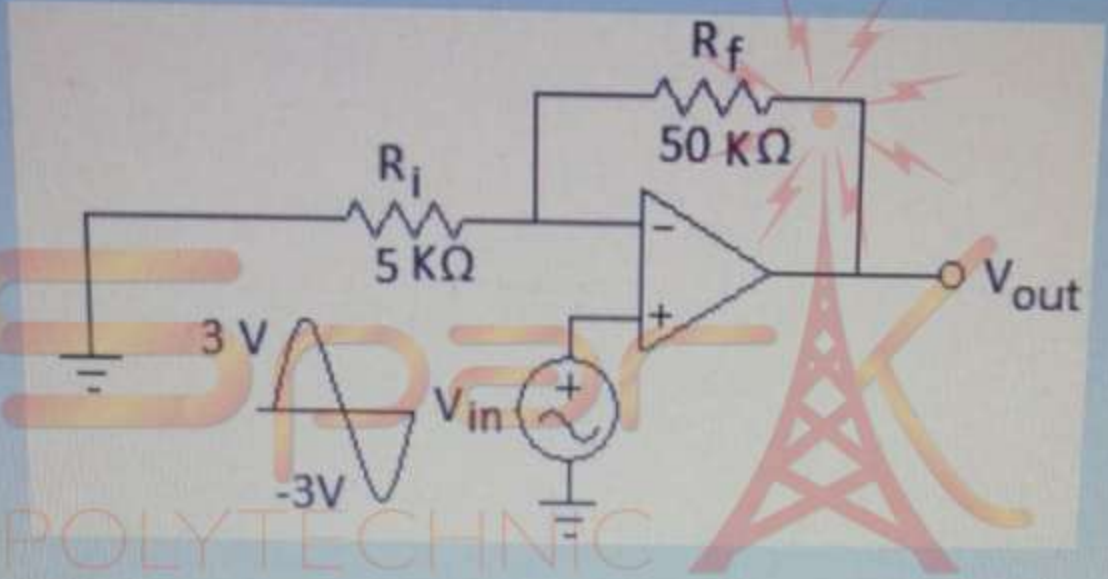
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RCUITS LAB-3

اسم الدائره في الشكل هي دائره



Select one:

- a. المكبر العاكس
- b. المنظم
- c. المرشح
- d. المكبر غير العاكس

D

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ND CIRCUITS LAB-3

Question 9

Not yet answered

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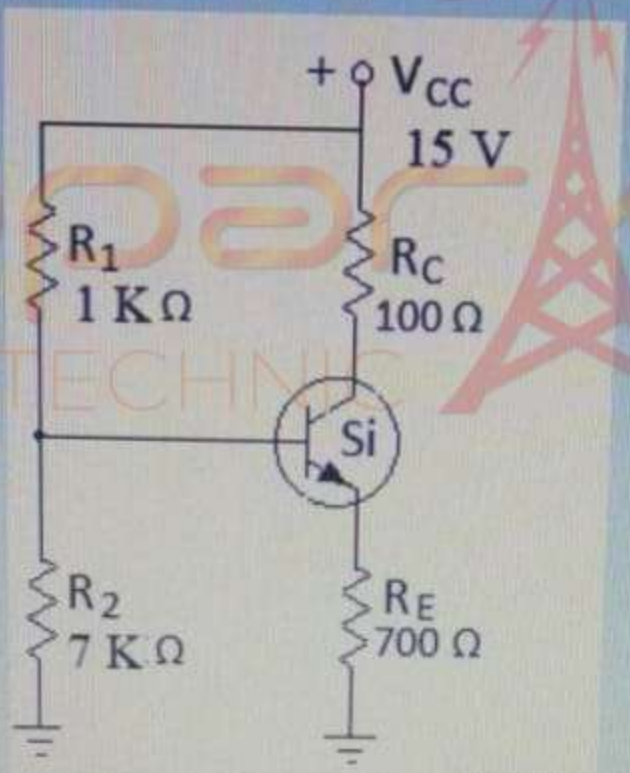
Flag question

إذا كان تيار المجمع يساوي

Ic=3 mA

فان جهد الباعث يساوي

Ve



Select one:

- a. 0 V
- b. 5.3 V
- c. 3.5 V
- d. 2.1 V

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400112- ELECTRONIC CIRCUITS LAB-3

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في دائره مقوم موجة كاملة إذا كانت قيمة

Vp(out)=5 V

فإن قيمة

V_{DC}

تساوي

Select one:

- a. 6.37 V
- b. 3.18 V
- c. 10 V
- d. 1.41 V

B

Next page

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2- ELECTRONIC DEVICES AND CIRCUITS LAB

Question 15

Not yet answered

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Flag question

لترانزستور تأثير المجال

إذا علمت ان

I_{DSS}=12 mA, V_{GS}off= -10 V

فإن قيمة تيار المصرف عند

V_{GS}=0 V

Select one:

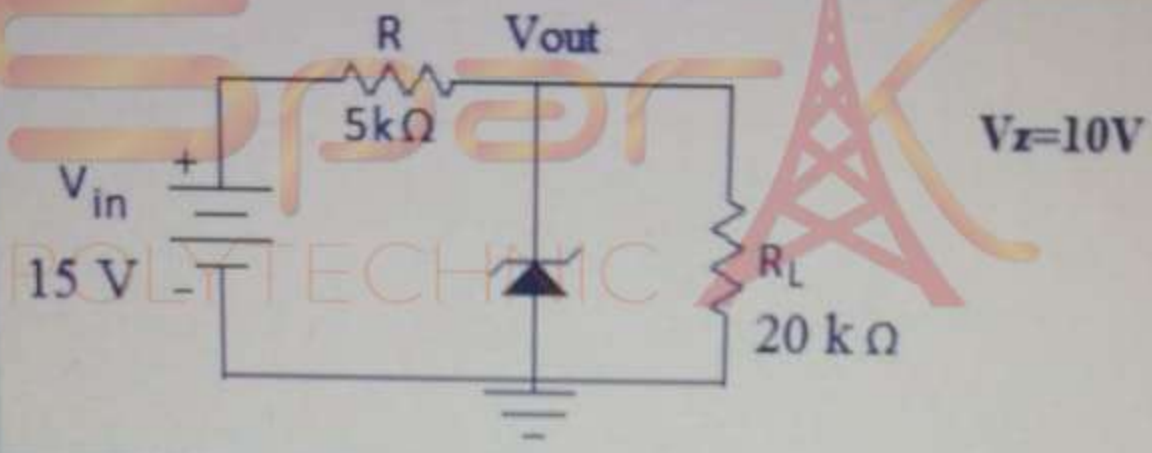
- a. 12 mA
- b. 4 mA
- c. 0
- d. 25 mA

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RCUITS LAB-3

احسب تيار الحمل

I_L



Select one:

- a. 1mA
- b. 0.5mA
- c. 0mA
- d. 2mA

B

Next page

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00112- ELECTRONIC
RCUITS LAB-3

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جهاز راسم اشارة يستخدم لقياس

Select one:

- a. الزمن و المقاومة
- b. الجهد و الزمن الدوري
- c. التيار و الجهد
- d. التيار و المقاومة

B

Next page

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00112- ELECTRONIC RCUITS LAB-3

في دائره منظم زينر أذا كانت

V(no loud)=8V, V(loud)= 7.9V

فان نسبة تنظيم الحمل تساوي

Select one:

- a. 4.67%
- b. 1.27%
- c. 0.7%
- d. 10%

B

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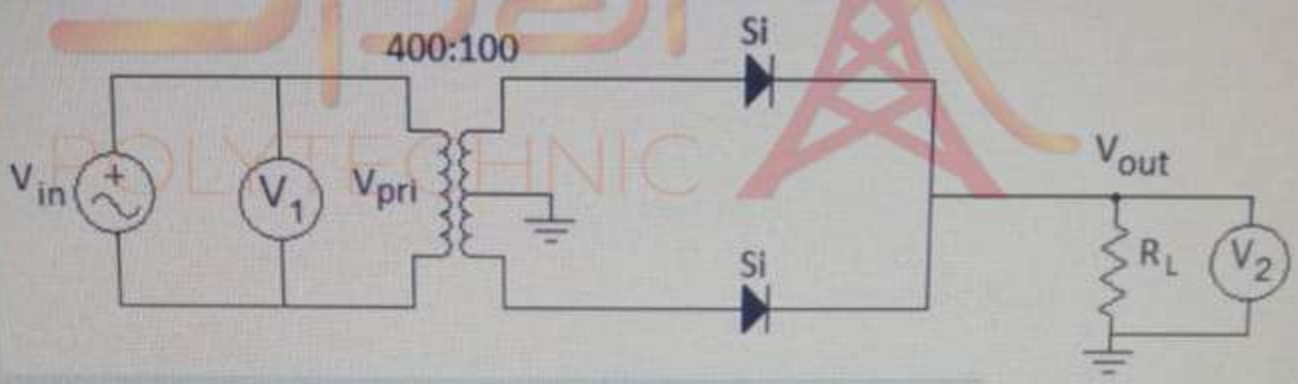
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0112- ELECTRONIC CUITS LAB-3

Vrms(in)=20V

احسب

Vp(pri)



Select one:

- a. 30.14 V
- b. 20 V
- c. 14.14 V
- d. 28.5 V

Next page

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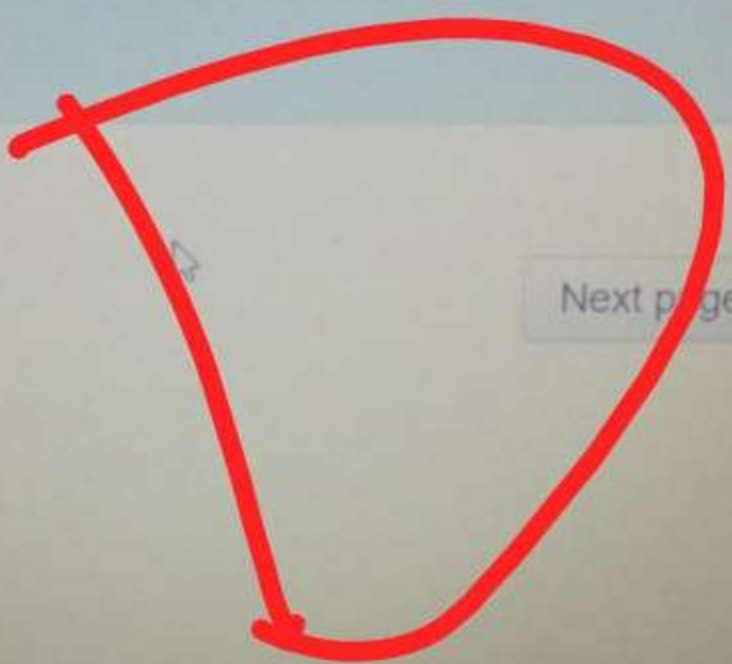
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0112- ELECTRONIC
CUITS LAB-3

عند توصيل القطب الموجب لمصدر الجهد مع الطرف المهبط للديود و الطرف السالب للمصدر مع المصعد, فإن مقاومة diode تكون

Select one:

- a. 0.7 Ω
- b. 0Ω
- c. منخفضة
- d. عالية



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400112- ELECTRONIC CIRCUITS LAB-3

5

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estion

عند استخدام المرشح يقل معامل التموج عند:

Select one:

- a. تم استخدام تقويم نصف الموجة
- b. تقليل قيم C و R
- c. تقليل جهد التموج
- d. تقليل قيمة التردد

Finish attempt ...

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400112- ELECTRONIC CIRCUITS LAB-3

14

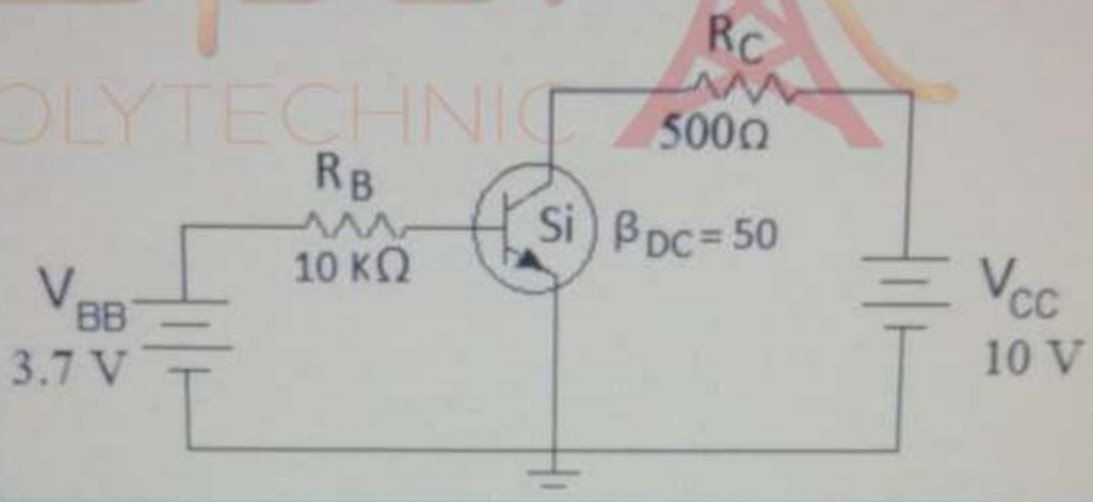
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إذا علمت ان الترانزستور مصنوع من السيليكون اوجد قيمة تيار القاعده

I_B



Select one:

- a. 0.47 mA
- b. 4 A
- c. 2.4 mA
- d. 0.3 mA

Next page

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0400112- ELECTRONIC CIRCUITS LAB-3

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ag question

عند فحص ترانزستور

BJT npn

إذا تم وصل القطب الموجب لجهاز القياس مع الباعث و الطرف السالب مع المجمع فإن قراءة الجهاز تكون

Select one:

- a. OL
- b. 0.7 V
- c. - 0.7 V
- d. 0

Spark POLYTECHNIC

A

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20400112- ELECTRONIC
CIRCUITS LAB-3

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Flag question

لترانزستور تأثير المجال إذا علمت ان

I_{DSS} = 12 mA, V_{GS}off = - 10 V

فان قيمة جهد الانحصار يساوي

(Vp)

Select one:

- a. -5 V
- b. 0
- c. -10 V
- d. 10 V



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00112- ELECTRONIC RCUITS LAB-3

في دائره مكبر عمليات عاكس إذا علمت ان قيم

Rf=20 kΩ , Ri=5 kΩ

فان معامل التكبير يساوي

Av

Select one:

- a. - 4
- b. 5
- c. 4
- d. - 5

Next page

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In Class B push – pull amplifier both transistor are biased at

اختر احد الخيارات

a. linear operation ○

b. maximum power ◉

c. saturation ○

d. cutoff ○

SPARK POLYTECHNIC

?Which is greater, the breakdown voltage or the barrier potential .

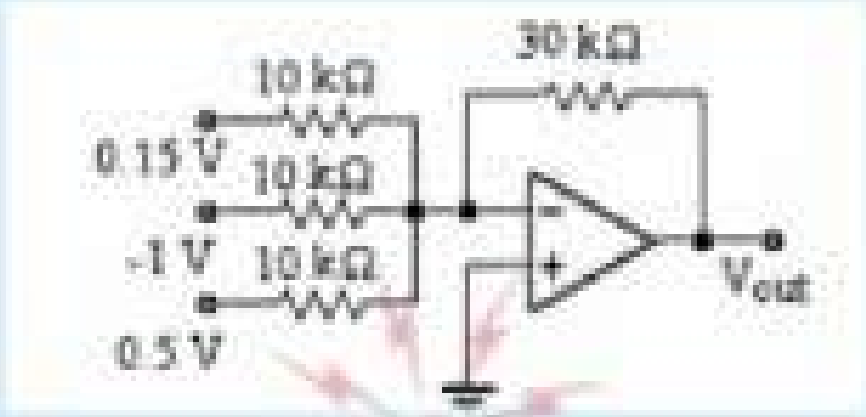
اختر احد الخيارات

a. Breakdown voltage is always much less than the barrier potential ○

b. Breakdown voltage is always equal the barrier potential ◉

c. Breakdown voltage is always much greater than the barrier potential ○

d. none ○



A voltmeter placed from the inverting input to ground would read

اختر احد الخيارات

a. 2.775 v

b. -2.775 v

c. 0 v

d. -0.925 v

: In JFET, the channel is between

اختر أحد الخيارات

a. Gate and Drain

b. Has no channel

c. Source and Drain

d. Gate and Source

?What symptoms does an open R_C produce

اختر أحد الخيارات

a. If RC is open VCC= VCE

.b. If RC is open, VC is about 0.7 V less than VB

c. If RC is open VCC=0

d. If RC is open VCE=0

For a class A CE amplifier, the power dissipation P_{DQ} is

اختر احد الخيارات

a. is zero with no input signal ○

b. is maximum when there is no input signal ○

c. increases when the peak to peak load voltage increases ○

d. is maximum when the transistor is driven to cutoff ◉

?In a zener diode regulator, what value of load resistance results in the maximum zener current

اختر أحد الخيارات

a. SHORT (O Ω) ○

b. 100Ω ○

c. 10K ○

d. An infinite resistance (open ◉

Spark POLYTECHNIC

If the load resistance connected to a filtered power supply is decreased, what happens to the ripple voltage

اختر أحد الخيارات

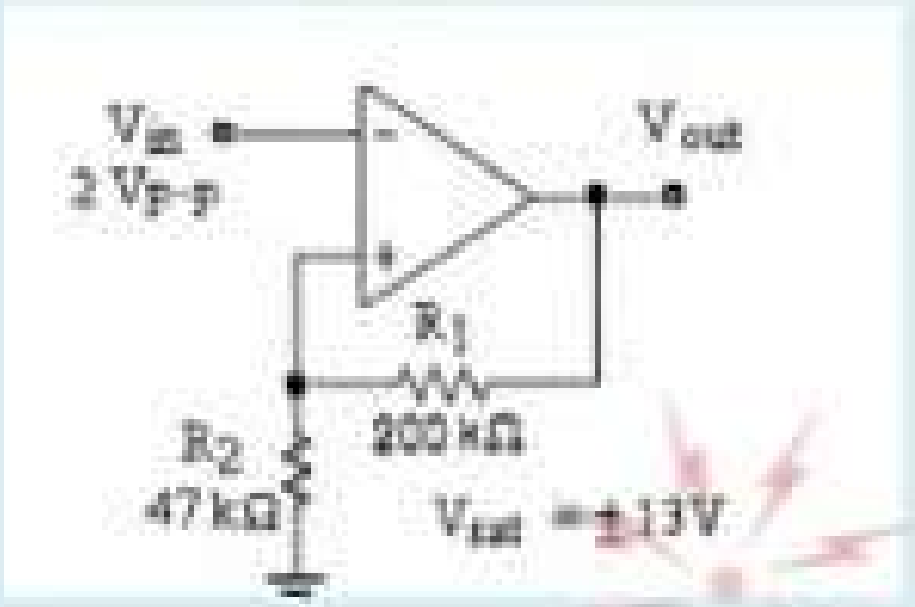
a. If the load resistance connected to a filtered power supply is decreased, what happens to the ripple voltage ○

b. not affected ◉

c. If the load resistance connected to a filtered power supply is increased, what happens to the ripple voltage ○

d. the same ○

Spark POLYTECHNIC



The output voltage with the inputs as shown

اختر واحد للفيزات

a. -Vsat

b. 17.06 Vp-p

c. +Vsat

d. 26 Vp-p

In an op-amp integrator, the current through the input resistor flows into the:

اختر أحد الخيارات

a. inverting input ○

b. noninverting input ○

c. bypass capacitor ○

d. feedback capacitor ◉

Spark POLYTECHNIC

What causes the ripple voltage on the output of a capacitor-input filter

اختر أحد الخيارات

a. The ripple voltage is caused by the slight discharging of the capacitor through the load resistor ○

b. none ○

c. The ripple voltage is caused by the slight charging and discharging of the capacitor through the load resistor ○

d. The ripple voltage is caused by the slight charging of the capacitor through the load resistor ◉

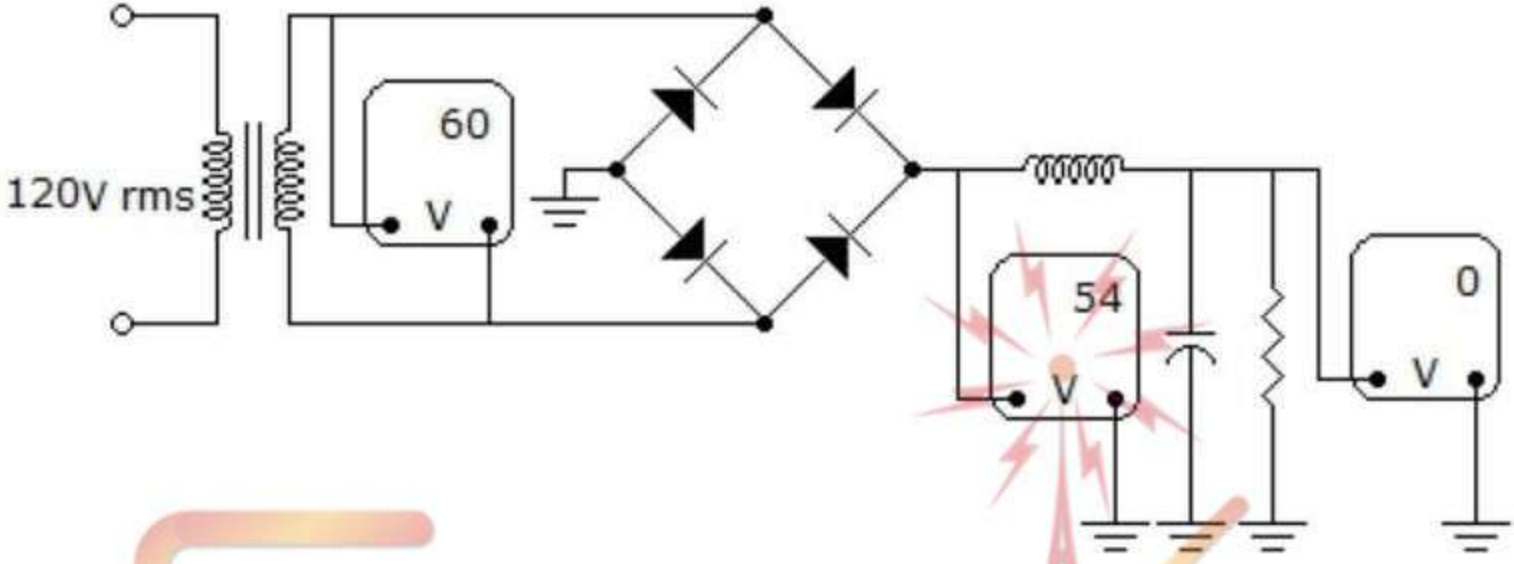
Spark POLYTECHNIC

When operating the transistor, a dc biasing is necessary to establish the proper region of operation for ac amplification

Select one:

- a. True
- b. False

52. Refer to the figure given below. If the voltmeter across the transformer reads 0 V, the probable trouble, if any, would be



A. one of the diodes is open.

B. a diode is shorted.

C. an open transformer secondary.

D. the filter capacitor is shorted.

E. no trouble exists.

Answer: Option C

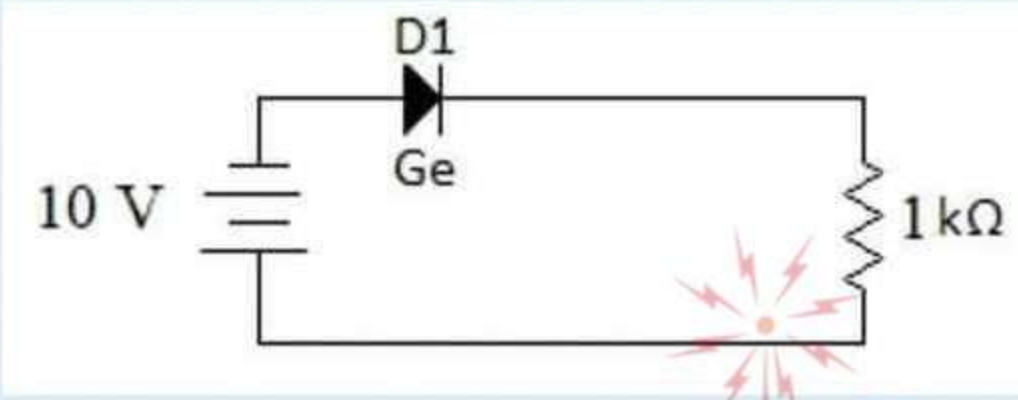
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For the positive clamper circuit and the input signal is square wave (+4 v, -5 v) Which of the following is true for the outp

Select one

- a. The output waveform is a square wave with 0V at peak compared to 0V
- b. The output wave is a square wave with 5V at peak compared to 0V
- c. The output waveform is a square wave with 4V at peak compared to 0V
- d. The output waveform is a straight line with the value of peak compared to 0V

The value of VR is:



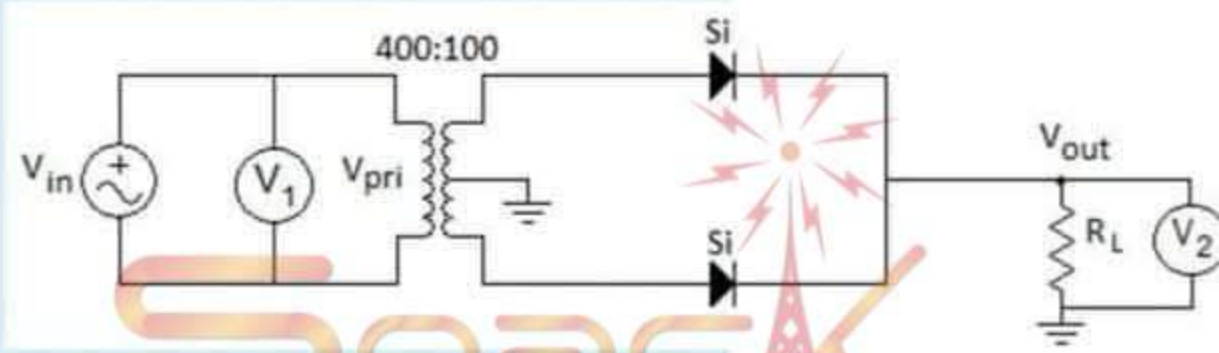
Select one:

- a. 10 V
- b. 9.3 V
- c. 0 V
- d. 9.7 V

Spark POLYTECHNIC
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Vp(sec)=20V

Calculate the value of Vp(out)



Select one:

- a. 5 V
- b. 14.14V
- c. 28.28V
- d. 9.3 V

Spark POLYTECHNIC

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What type of diode circuit is used to add or restore a dc level to an electrical signal?

Select one:

- a. IC voltage regulator
- b. none of the mentioned
- c. clamper
- d. clipper or limiter

Te setup Common Drain you need the following

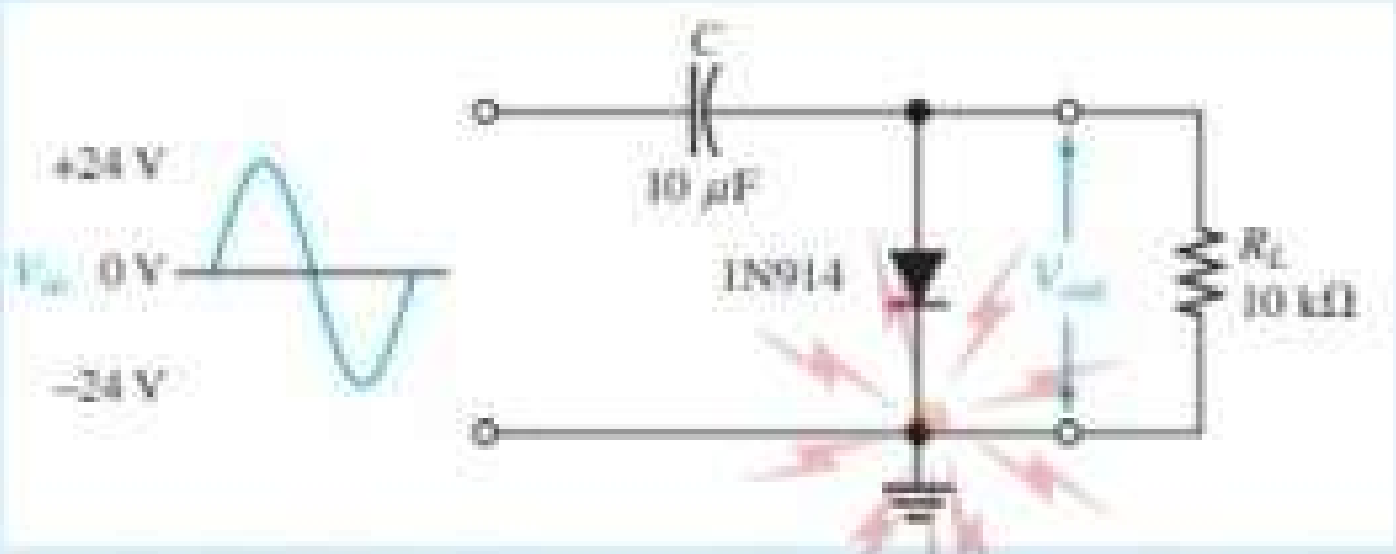
اختر أحد الخيارات

a. Either Ac or DC

b. Both DC and Ac

c. Only Ac voltage

d. Only DC voltage



What is the output voltage that you would expect to observe across RL

اختر أحد الخيارات

a. 0.7--47.3 V ○

b. +(24 V + 0.7 V) ○

c. 0 -48 v ◉

d. -(24 V - 0.7 V) ○

_____ A transistor can be checked using a(n)

اختر أحد الخيارات

a. curve tracer

b. ohmmeter

c. digital meter

d. Any of the above

Spark POLYTECHNIC

How do you determine when a transistor is saturated? When a transistor is in cutoff .

اختر أحد الخيارات

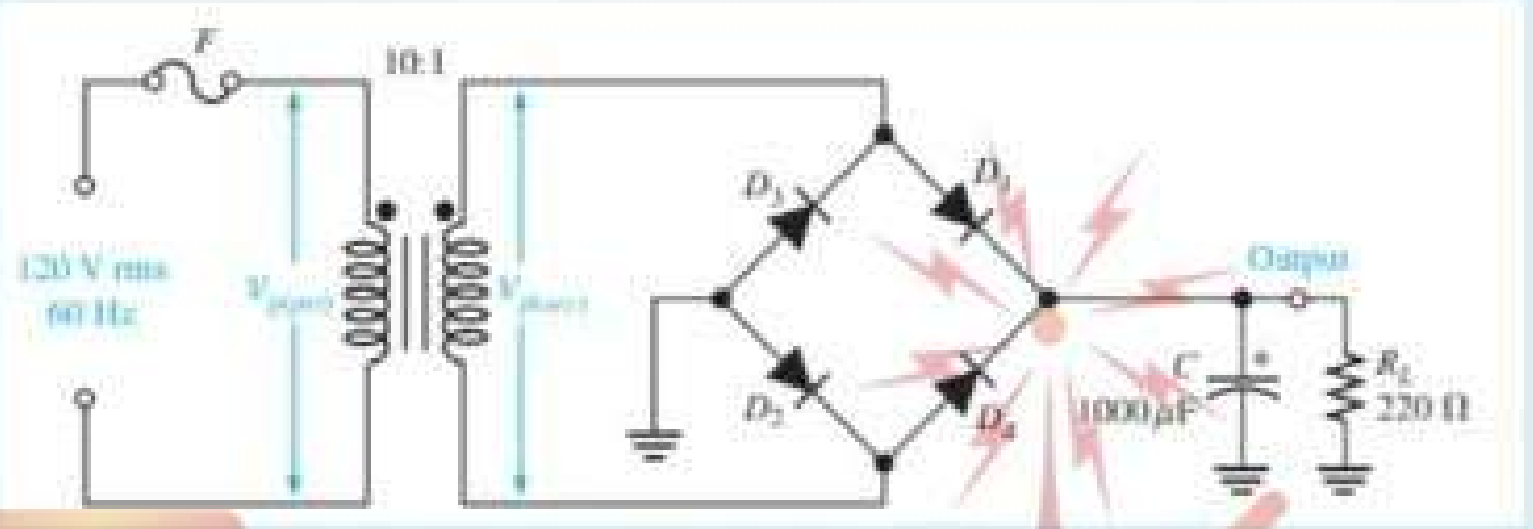
a. $V_{CESAT} = V_{CC}$,, $V_{CECUT} = 0V$

b. $V_{CE\ SAT} = V_{CC}$,,, $V_{CE\ CUT} = V_{CESAT}$

c. $V_{CECUT} = VCE_{SAT}$

d. $V_{CESAT} = 0\ V$. .....$V_{CECUT} = V_{CC}$

Spec POLYTECHNIC



Determine the ripple factor for the filtered bridge rectifier with a load as indicated in Figure

اختر أحد الخيارات

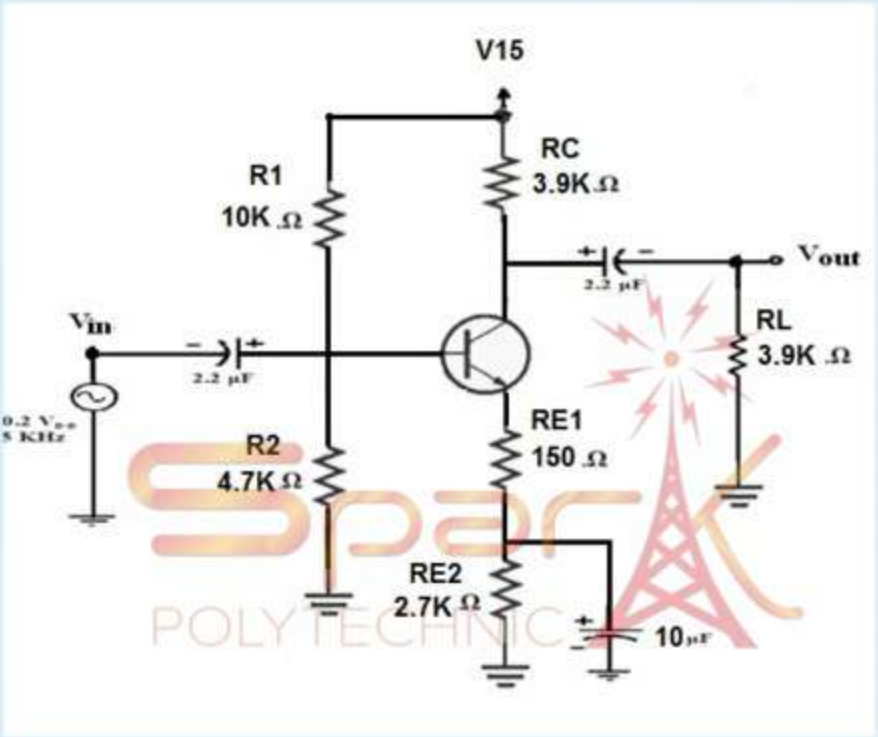
a. 2.3

b. 1.7

c. 0.227

d. 0.39

For the circuit shown find V_{CE}



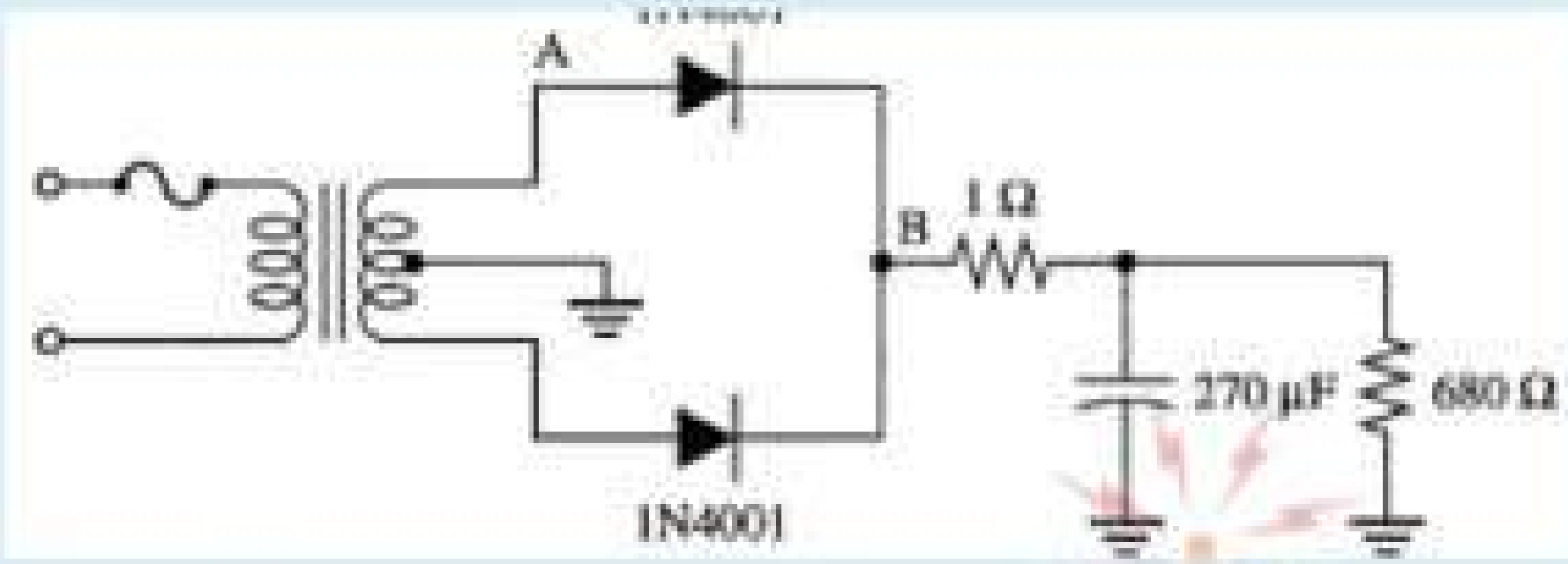
اختر أحد الخيارات

a. 0.7 V

b. 5.28 V

c. 10.9 V

d. 9.38 V



. The voltage at point A with respect to ground Is 12.7 v

=PIV

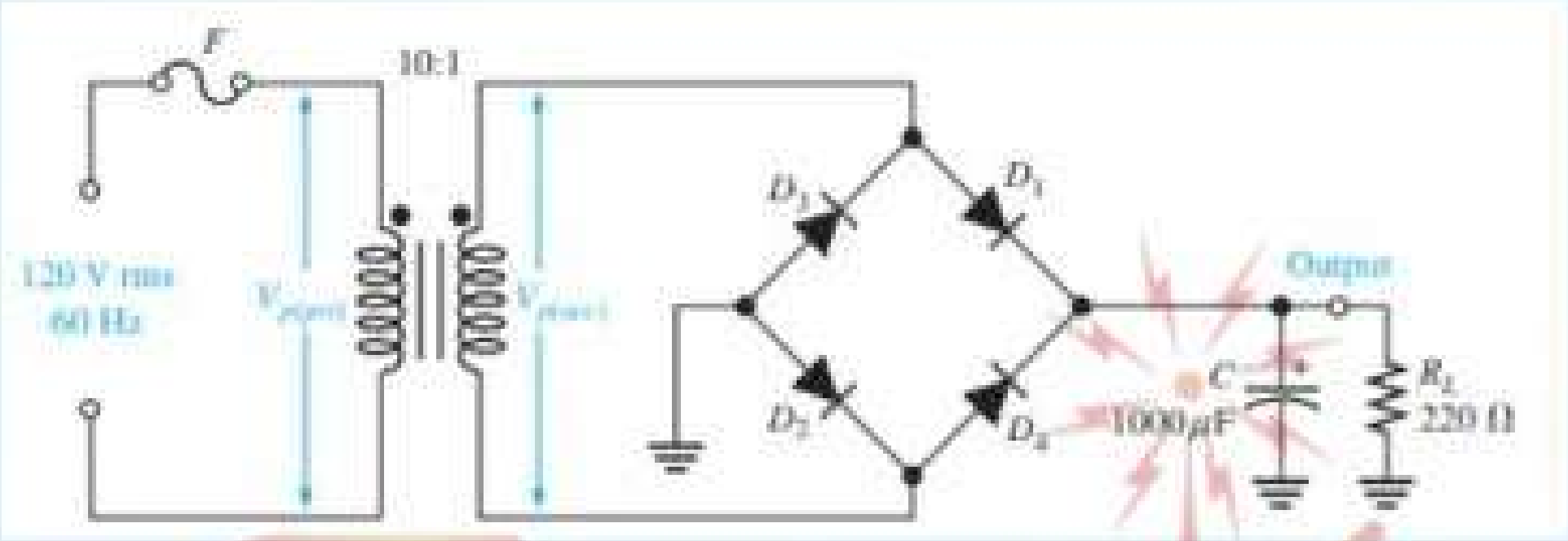
اختر احد الخيارات

a. 24.7 V ○

b. 12.7 V ◉

c. 36.7V ○

d. 6.7 V ○



The approximate dc value of the output voltage is

اختر احد الخيارات

a. 15.6 V ○

b. 17 V ○

c. 15.3 V ◉

d. 1.56V ○

Spark POLYTECHNIC

In a voltage-divider biased npn transistor circuit, you measure VCC at the collector and an emitter voltage 0.7 V less than the base voltage. Is the .transistor functioning in cutoff, or is R_E open?

اختر أحد الخيارات

a. none ○

b. RE is open because the BE junction of the transistor is still reverse -biased ○

c. RE is open because the BE junction of the transistor is still forward-biased ○

d. RE is short because the BE junction of the transistor is still forward-biased ◉

What is the maximum voltage across an unbiased positive silicon diode limiter during the positive alternation of the input voltage .

اختر أحد الخيارات

a. 5 V appears across the diode ○

b. 0.7 V appears across the diode ◉

c. 0.5 V appears across the diode ○

d. PIV ○

Spark POLYTECHNIC

To limit the output voltage of a positive limiter to 5 V when a 10 V peak input is applied, , what value must the bias voltage be

اختر أحد الخيارات

a. 0V ○

b. The bias voltage must be 0.7 V ○

c. The bias voltage must be 5 V + 0.7 V = 5.7 V ○

d. The bias voltage must be 5 V - 0.7 V = 4.3 V ◉

What is the difference between input (line) regulation and load regulat

اختر أحد الخيارات

a. Input regulation is a measure of the variation in output voltage over a range of output voltages. Load regulation is a measure of the variation in output voltage over a range of load current values. ○

b. Input regulation is a measure of the variation in output voltage over a range of input voltages. Load regulation is a measure of the variation in output voltage over a range of load current values. ○

c. Input regulation is a measure of the variation in input voltage over a range of input voltages. Load regulation is a measure of the variation in input voltage over a range of load current values. ○

d. all of above ◉

?At what point on the input cycle does the PIV occur .

اختر أحد الخيارات

a. PIV across the diode occurs at the peak of the input when the diode is reversed biased ○

.b ○

$PIV = V_D$

c. none ○

d. PIV across the diode occurs at the peak of the output when the diode is reversed biased ◉

Spar POLYTECHNIC

If the input to a comparator is a sine wave, the output is a:

اختر احد الميزات

a. ramp voltage ○

b. sawtooth wave ○

c. sine wave ◉

d. rectangular wave ○

Spa POLYTECHNIC

Negative feedback

Select one:

- a. Controls the gain
- b. Limits the affect of the DC
- c. None is correct
- d. Effects the phase

Spark

ة ضوئيا بـ CamScanner

When the regulation by a Zener diode is with a varying input voltage, what happens to the voltage drop across the resistance?

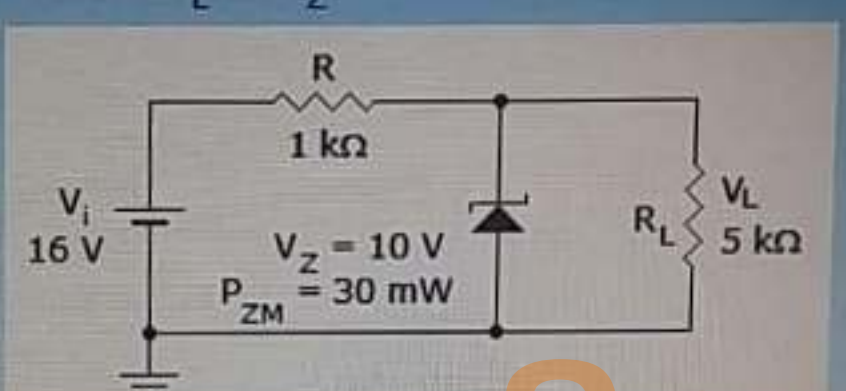
Select one:

Spark

- a. Decreases
- b. The variations depend on temperature
- c. Increases
- d. Has no effect on voltage

ة ضوئيا بـ CamScanner

Calculate I_L and I_Z



Spark

Select one:

- a. 4 mA, 2 mA
- b. 4 mA, 2 mA
- c. 2 mA, 4 mA
- d. 2 mA, 0 mA

CamScanner ـة ضوئيا بـ

N - JFET Terminals when tested , results are :

Select one:

- a. 0.7 V between Drain and Source
- b. Zero between Gate and Source
- c. Zero (short) between Drain and Source
- d. Zero between Gate and Drain

Spark

ة ضوئيا بـ CamScanner

In the active region, while the collector-base junction is _____-biased, the base-emitter is _____-biased.

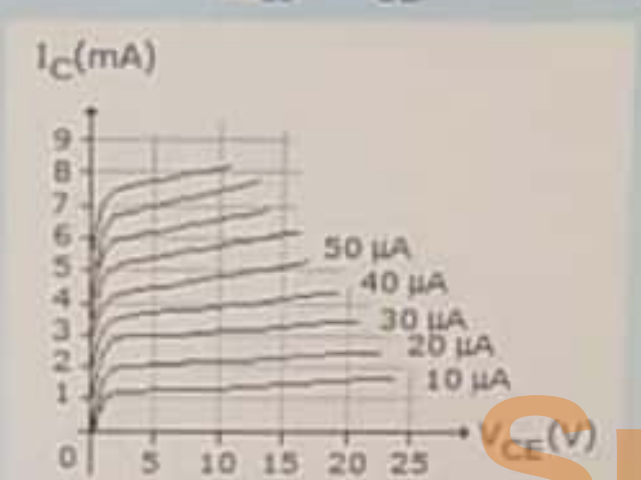
Select one:

- a. reverse, forward
- b. forward, reverse
- c. reverse, reverse
- d. forward, forward

Spark

ة ضوئيا بـ CamScanner

Calculate β_{dc} at $V_{CE} = 15$ V and $I_B = 30$ μA



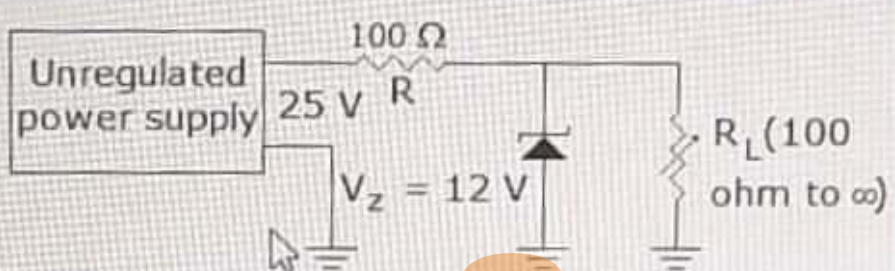
Spark

Select one:

- a. 116
- b. 110
- c. 50
- d. 100

ة ضوئيا بـ CamScanner

In figure The minimum and maximum load currents are



Spark

Select one

- a. 0 and 120 mA
- b. 0 and 60 mA
- c. 10 mA and 120 mA
- d. 10 mA and 60 mA

CamScanner ـة ضوئيا بـ

In a certain biased limiter, the bias voltage is 5 V, and the input is 10 V peak sine wave. If the positive terminal of the bias voltage is connected to the cathode of the diode, the maximum voltage of the anode is:

Select one:

- a. 0.7 V
- b. 5.7 V
- c. 5 V
- d. 10 V

Spark

ة ضوئيا بـ CamScanner

in which region are both the collector-base and base-emitter junctions forward-biased?

Select one:

- a. saturation
- b. All of the above
- c. cutoff
- d. Active

Spark

CamScanner ـة ضوئيا بـ

What range of resistor values would you get when checking a transistor for forward- and reverse-biased conditions by an ohmmeter?

Select one:

- a. 100 Ω to a few k Ω, exceeding 100 k Ω
- b. Exceeding 100 k Ω, exceeding 100 k Ω
- c. Exceeding 100 k Ω, 100 Ω to a few k Ω
- d. 100 Ω to a few k Ω, 100 Ω to a few k Ω

Spark

ة ضوئيا بـ CamScanner

In negative clamper, one of the following conditions should be satisfied for the sinusoidal signal of period T

Select one

- a. 4.3 v
- b. RC << T
- c. RC >>T
- d. RC =0.35 T
- e. RC =T

Spark

ة ضوئيا بـ CamScanner

For a clamping circuit , Sometimes an independent dc supply is also required to cause an:

Select one:

- a. additional frequency.
- b. additional V peak output.
- c. additional V peak input
- d. additional shift.

Spark

ة ضوئيا بـ CamScanner

The maximum value of collector current in a biased transistor is

Select one:

- a. I_{CSAT}
- b. $I_E - I_B$
- c. greater than I_E
- d. $I_B \beta_{dc}$

Spark

ة ضوئيا بـ CamScanner

For the voltage-divider-bias configuration, a common-emitter amplifier circuit, V_{CC} =15 V, R_1 =10 kΩ, R_2 = 4.7 kΩ, R_C = 3.9 kΩ, R_{E1} = 150 Ω, R_{E2} = 2.7 kΩ, R_L = 3.9 kΩ, V_i = 0.2 V_{P-P} (5 kHz), C_1 = C_2 =2.2 μF, Bypass capacitor C_3 = 10 μF . We use the Bypass capacitor in order to

Spark

Select one:

- I. no effect of the bypass capacitor
- II. increase the voltage gain
- III. decrease the voltage gain
- IV. remove the noise in the circuit

ة ضوئيا بـ CamScanner

For the voltage-divider-bias configuration, a common-emitter amplifier circuit V_{CC} =15 V, R_1 =10 kΩ, R_2 = 4.7 kΩ, R_C = 3.9 kΩ R_{E1} = 150 Ω, R_{E2} = 2.7 kΩ, R_L = 3.9 kΩ, V_i = 0.2 V_{p-p} (5 kHz), C_1 = C_2 =2.2 μF, Bypass capacitor C_3 = 10 μF Find V_E

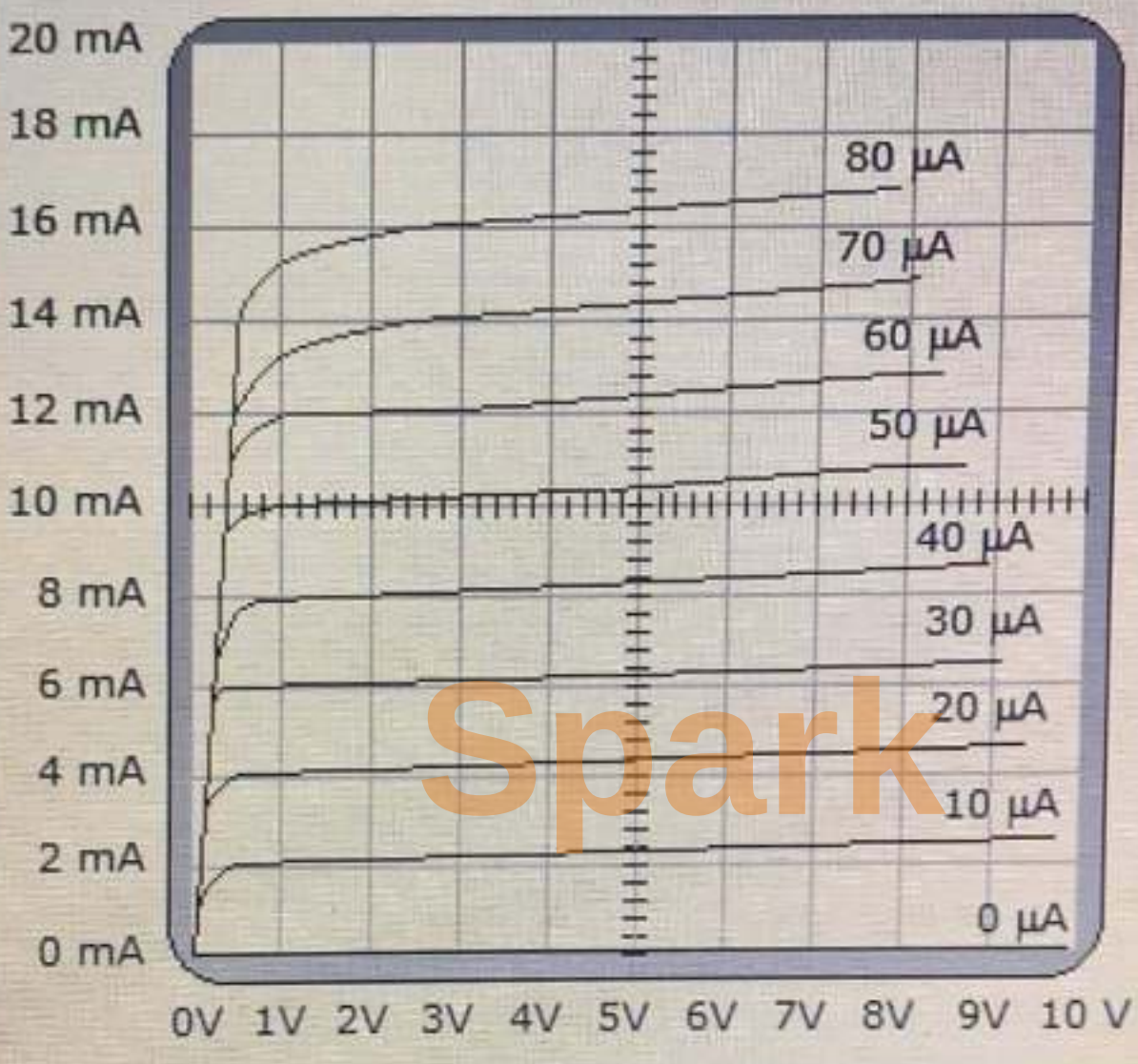
Select one

- I. 9.5 V
- II. 14.3 V
- III. 4.1 V
- IV. 0.7 V

Spark

CamScanner ـة ضوئيا بـ

Calculate β_{ac} for $I_C = 15$ mA and $V_{CE} = 5$ V.



Spark

Select one:

- a. 200
- b. 180
- c. 220
- d. None of the above

CamScanner ـة ضوئيا بـ

What does a reading of a large or small resistance in forward- and reverse-biased conditions indicate when checking a transistor using an ohmmeter?

Select one:

- a. faulty device
- b. Good device
- c. Bad ohmmeter
- d. None of the above

Spark

ة ضوئيا بـ CamScanner

A transistor can be checked using a(n) _____

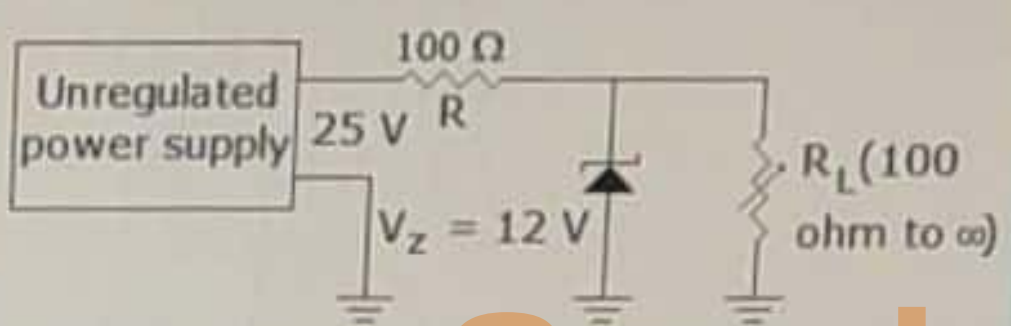
Select one:

- a. curve tracer
- b. digital meter
- c. Any of the above
- d. ohmmeter

Spark

ة ضوئيا بـ CamScanner

In figure The minimum and maximum load currents are



Spark

Select one:

- a. 10 mA and 60 mA
- b. 10 mA and 120 mA
- c. 0 and 120 mA
- d. 0 and 60 mA

CamScanner ـة ضوئيا بـ

The JFET is :

Select one:

- a. All is correct
- b. Voltage controlled device
- c. A unipolar device
- d. Operates with reverse - biased on junction

Spark

ة ضوئيا بـ CamScanner

For the positive clamper circuit and the input signal is square wave: (+4 v , -6 v) Which of the following is true for the output voltage?

Select one:

- a. The output waveform is a square wave with highest peak clamped to 4V
- b. The output waveform is a straight line with the value of output voltage equal to 10V
- c. The output waveform is a square wave with lowest peak clamped to 0V
- d. The output wave is a square wave with lowest peak clamped to -6V

Spark

CamScanner ـة ضوئيا بـ

Zener diodes are also known as

Select one:

- a. Breakdown diode
- b. None of the mentioned
- c. Voltage regulators
- d. Forward bias diode

Spark

ة ضوئيا بـ CamScanner

Which of the following is true about the resistance of a Zener diode?

Select one:

- a. It has an incremental resistance
- b. All of the mentioned
- c. The value of the resistance is the inverse of the slope of the i-v characteristics of the Zener diode
- d. It has dynamic resistance

Spark

CamScanner ـة ضوئيا بـ

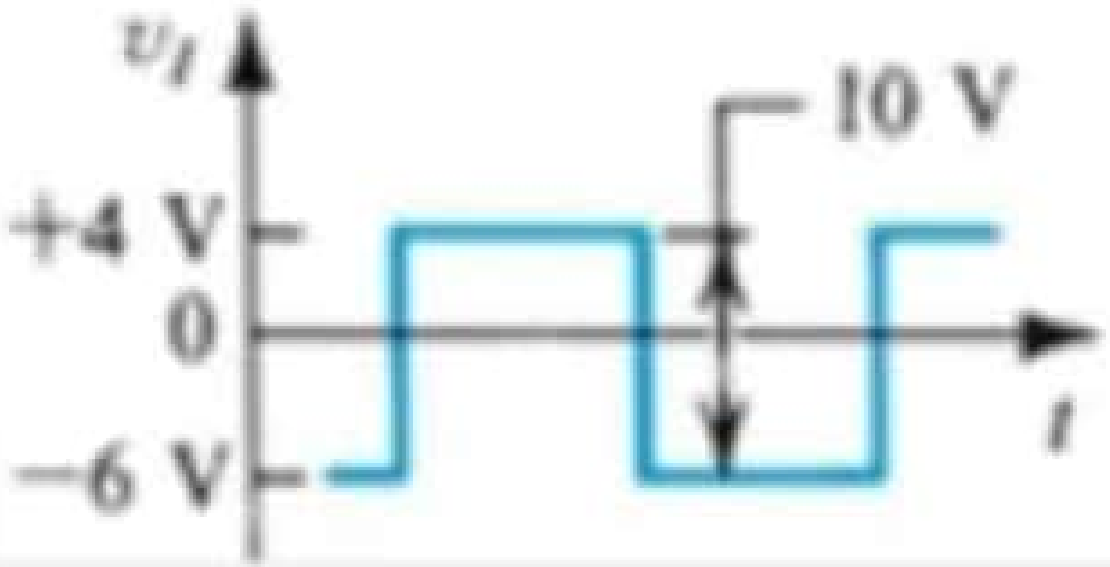
7

The graphical method of modeling a diode characteristics is based on (2 Points)

- Iteration method
- Constant voltage drop method
- Small signal approximation
- Exponential method

Spark

CamScanner ـة ضوئيا بـ



The output waveform is a square wave with lowest peak clamped to 0V

Spark

The output wave is a square wave with lowest peak clamped to -6V

The output waveform is a square wave with highest peak clamped to 4V

The output waveform is a straight line with the value of output voltage equal to 10V

CamScanner

3

For the same secondary voltage, the output voltage from a centre-tap rectifier is ... that of bridge rectifier.
(2 Points)

- twice
- thrice
- four time
- one-half

Spark

CamScanner ـة ضوئيا بـ

Hi Al-Mothana, when you submit this form, the owner will be able to see your name and email address.

1



The circuit inside the contains only resistor and diodes. The terminal voltage vo is connected to some point in the circuit inside the box. The largest and smallest possible value of vo most nearly to is respectively
(2 Points)

- 15 V, 6 V
- 24 V, 0 V
- 24 V, 6 V
- 15 V, 9 V

Spark

CamScanner ـة ضوئيا بـ

6

DC power should be connected to forward bias a diode as follows: (2 Points)

- anode. + cathode
- cathode – anode
- + anode – cathode
- + cathode + anode

Spark

ة ضوئيا بـ CamScanner

2

The Vz you used in the experiment was:
(2 Points)

- 6V
- 6.1V
- 6.2V
- 6.3 V

Spark

ة ضوئيا بـ CamScanner

8

The graphical method of modeling a diode characteristics is based on
(2 نقطة)

Spark

- Iteration method
- Constant voltage drop method
- Small signal approximation
- Exponential method

ة ضوئيا بـ CamScanner

11

Which of the following is method to model a diode's forward characteristics? (2 Points)

- Iteration method
- Graphical method
- Constant-voltage drop model
- OAll of the mentionedption 4

Submit

Spark

ة ضوئيا بـ CamScanner

1

In a bridge full wave rectifier, the input sine wave is 250sin100t. The output ripple frequency will be_____
(2 Points)

- 50Hz
- 200Hz
- 100Hz
- 25Hz

Spark

ة ضوئيا بـ CamScanner

1

:The Vz you used in the experiment was

(2 نقطة)

Spark

6V

6.1V

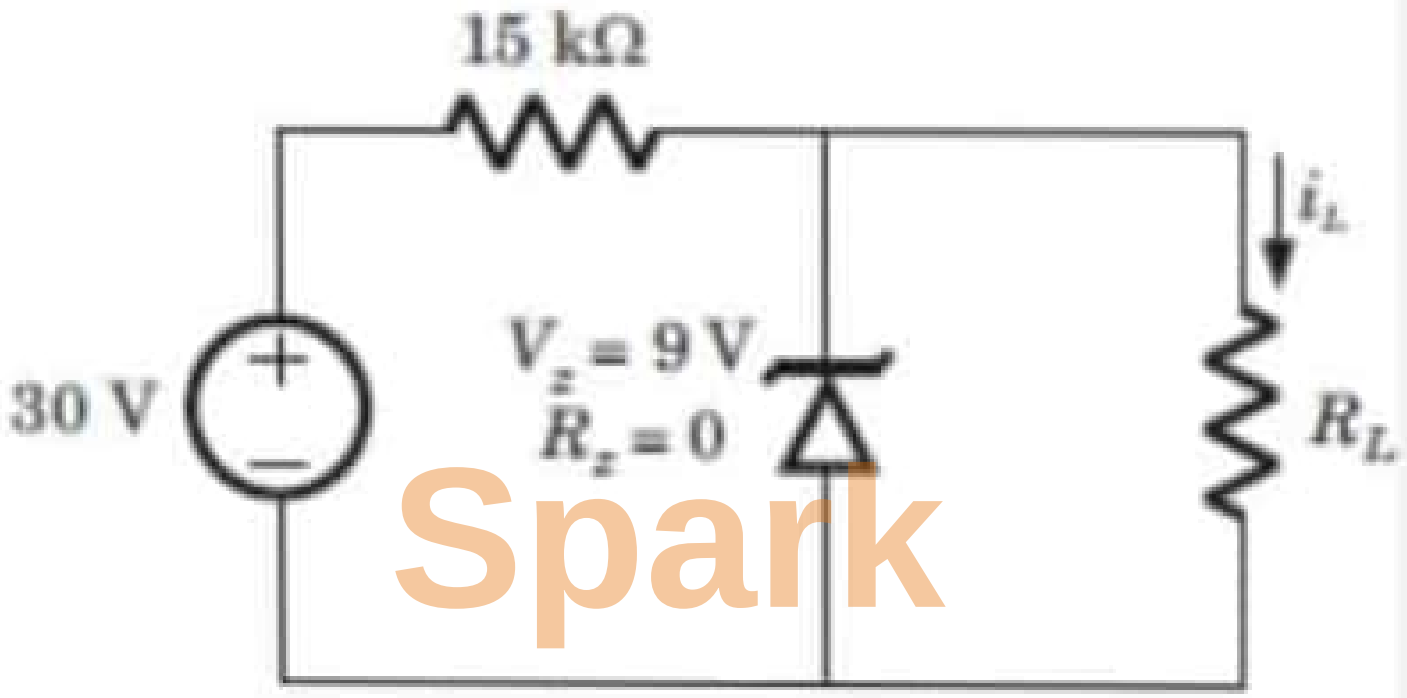
6.2V

V 6.3

ة ضوئيا بـ CamScanner

3

The maximum load current that can be drawn is: (2 نقطة)



Spark

- mA 1.4 ●
- mA 2.3 ○
- mA 1.8 ○
- mA 2.5 ○

CamScanner بـ صوتيا

8

For the same secondary voltage, the output voltage from a centre-tap rectifier is ..... than that of bridge rectifier
(0/2 Points)

Spark

- twice
- thrice
- four time
- one-half ✓

CamScanner ـة ضوئيا بـ

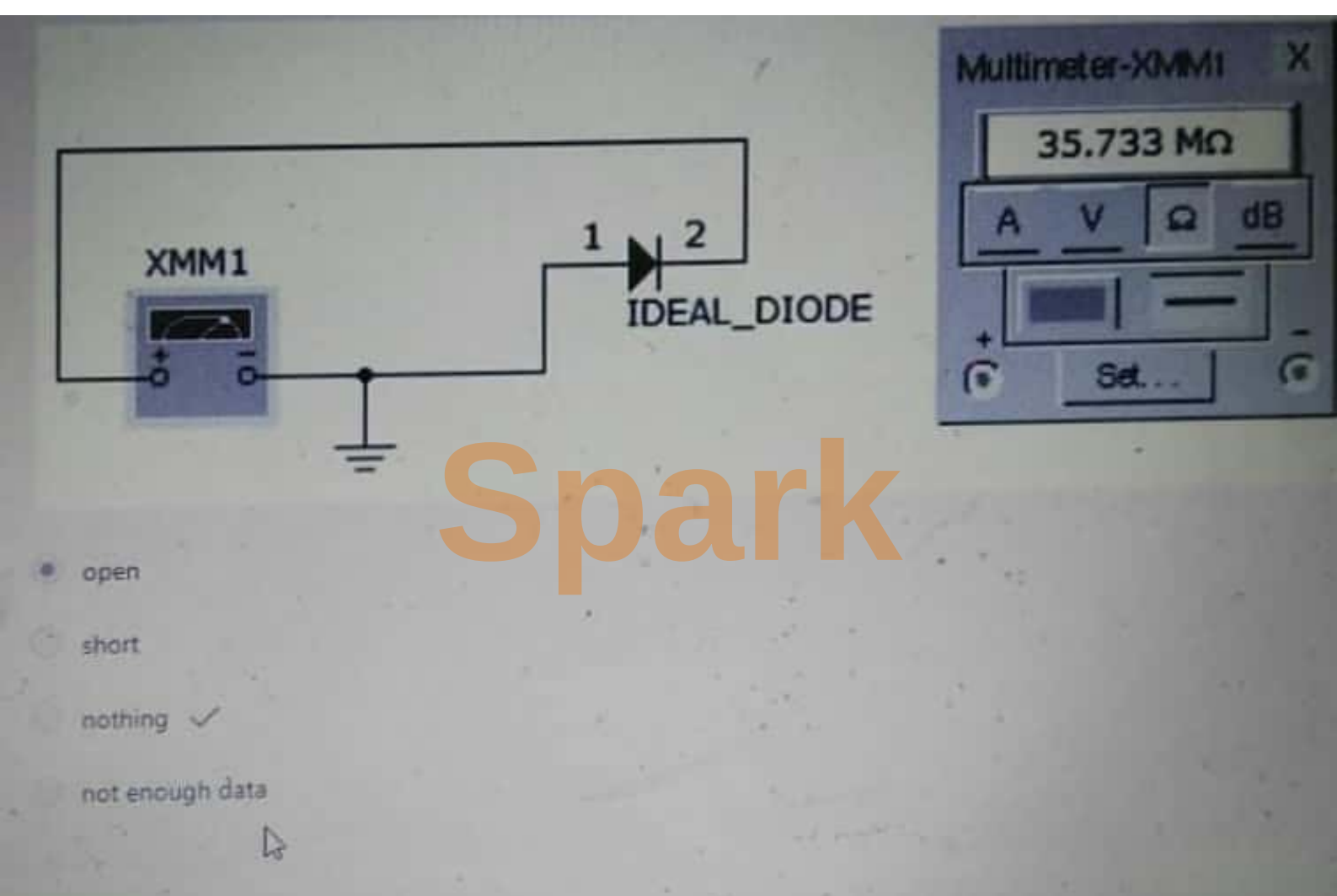
7

In a bridge full wave rectifier, the input sine wave is 250sin100t. The output ripple frequency will be_____
(0/2 Points)

Spark

- 50Hz
- 200Hz
- 100Hz ✓
- 25Hz

ة ضوئيا بـ CamScanner



- open
- short
- nothing ✓
- not enough data

ة ضوئيا بـ CamScanner

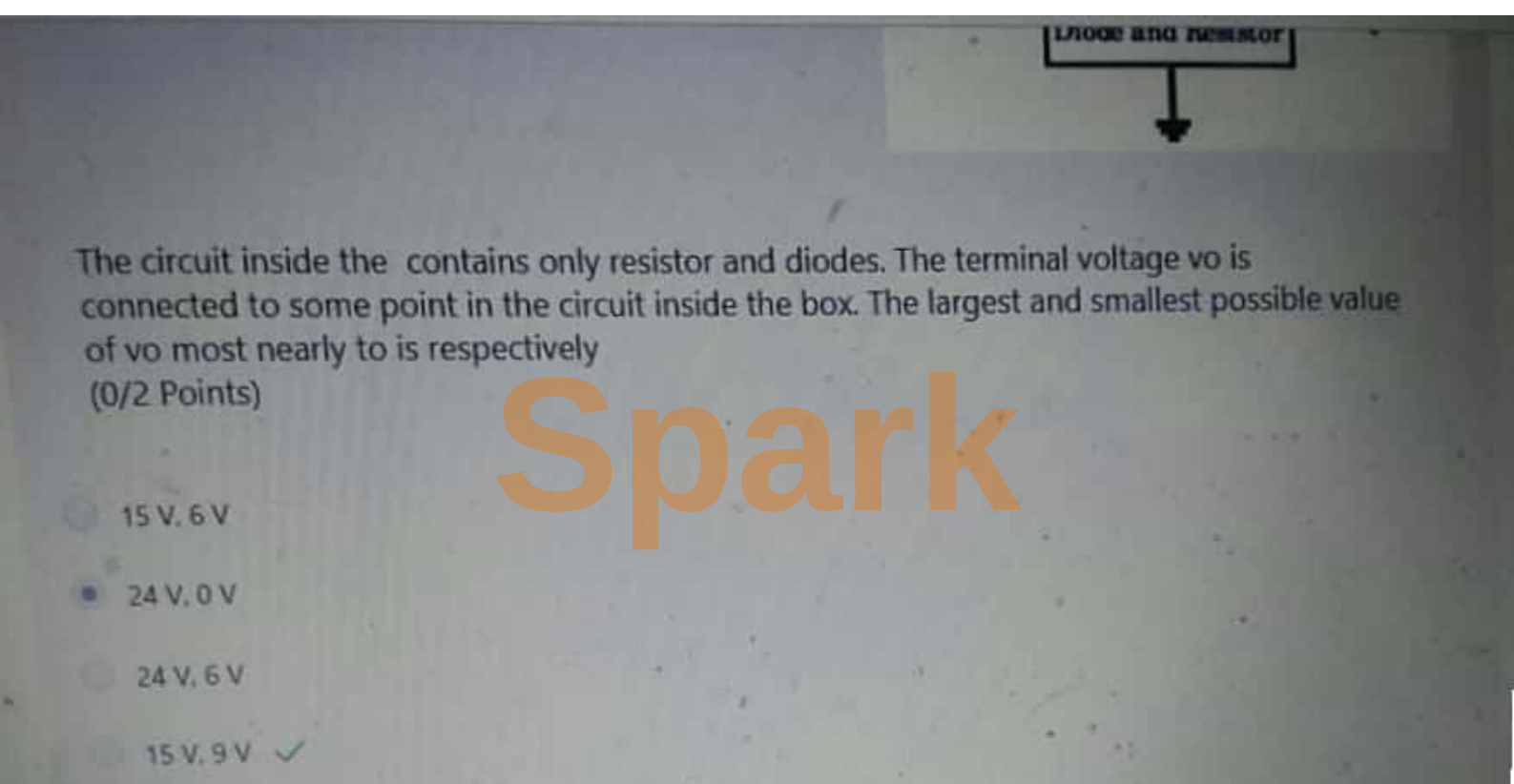
9

In Zener diode, for currents greater than the knee current, the *i-v* curve is almost
(2 Points)

- Almost a straight line parallel to y-axis
- Almost a straight line parallel to x-axis
- Equally inclined to both the axes with a positive slope
- Equally inclined to both the axes with a negative slope

Spark

CamScanner ـة ضوئيا بـ



The circuit inside the contains only resistor and diodes. The terminal voltage vo is connected to some point in the circuit inside the box. The largest and smallest possible value of vo most nearly to is respectively
(0/2 Points)

Spark

- 15 V, 6 V
- 24 V, 0 V
- 24 V, 6 V
- 15 V, 9 V ✓

CamScanner ـة ضوئيا بـ

4

The graphical method of modeling a diode characteristics is based on
(0/2 Points)

- Iteration method
- Constant voltage drop method
- Small signal approximation
- Exponential method ✓

Spark

ة ضوئيا بـ CamScanner

A transistor can be checked using a(n) _____

Select one:

- a. ohmmeter
- b. digital meter
- c. curve tracer
- d. Any of the above

Spark

ة ضوئيا بـ CamScanner

For the voltage-divider-bias configuration, a common-emitter amplifier circuit, V_{CC} =15 V, R_1 =10 kΩ, R_2 = 4.7 kΩ, R_C = 3.9 kΩ, R_{E1} = 150 Ω, R_{E2} = 2.7 kΩ, R_L = 3.9 kΩ, V_i = 0.2 V_{p-p} (5 kHz), C_1 = C_2 =2.2 μF, Bypass capacitor C_3 = 10 μF. Find voltage gain from base to collector in normal conditions

Select one:

- a. 1
- b. 0.88
- c. 23.2
- d. 11.6

Spark

ة ضوئيا بـ CamScanner

What type of diode circuit is used to add or restore a dc level to an electrical signal?

Select one:

- a. clipper or limiter
- b. clamper
- c. IC voltage regulator
- d. none of the mentioned

Spark

ة ضوئيا بـ CamScanner

A properly functioning diode in forward biased will produce a reading of

Select one:

- a. 0.5-0.9 V
- b. 0.2-0.7 V
- c. 0.5-5 V
- d. 0.5-9 V

Spark

ة ضوئيا بـ CamScanner

For the voltage-divider-bias configuration, a common-emitter amplifier circuit V_{CC} = 15 V, R_1 = 10 kΩ, R_2 = 4.7 kΩ, R_C = 3.9 kΩ, R_{E1} = 150 Ω, R_{E2} = 2.7 kΩ, R_L = 3.9 kΩ, V_i = 0.2 V_{p-p} (5 kHz), C_1 = C_2 = 2.2 μF, Bypass capacitor C_3 = 10 μF. Find V_E

Select one:

- a. 15 V
- b. 0.7 V
- c. 10.2V
- d. 4.8 V

Spark

ة ضوئيا بـ CamScanner

Diode clamper is:

Select one:

- a. Change the voltage level of an ac signal by adding a ac voltage
- b. Change the voltage level of an dc signal by adding a dc voltage
- c. Change the voltage level of an dc signal by adding a ac voltage
- d. Change the voltage level of an ac signal by adding a dc voltage

Spark

ة ضوئيا بـ CamScanner

At cutoff, the JFET channel is :

Select one:

- a. Extremely narraw
- b. Comletely closed by the depletion region
- c. At its widest point
- d. Reverse - biased

Spark

ة ضوئيا بـ CamScanner

For the voltage-divider-bias configuration, a common-emitter amplifier circuit V_{CC} = 15 V, R_1 = 10 kΩ, R_2 = 4.7 kΩ, R_C = 3.9 kΩ, R_{E1} = 150 Ω, R_{E2} = 2.7 kΩ, R_L = 3.9 kΩ, V_i = 0.2 V_{p-p} (5 kHz), C_1 = C_2 = 2.2 μF, Bypass capacitor C_3 = 10 μF. Find I_E

Select one:

- a. 2.73 mA
- b. 1.44 mA
- c. 1.95 mA
- d. 3.2 mA

Spark

CamScanner ـة ضوئيا بـ

When the regulation by a Zener diode is with a varying input voltage, what happens to the voltage drop across the resistance?

Select one:

- a. The variations depend on temperature
- b. Decreases
- c. Increases
- d. Has no effect on voltage

Spark

ة ضوئيا بـ CamScanner

Ripple frequency of a half wave rectifier is

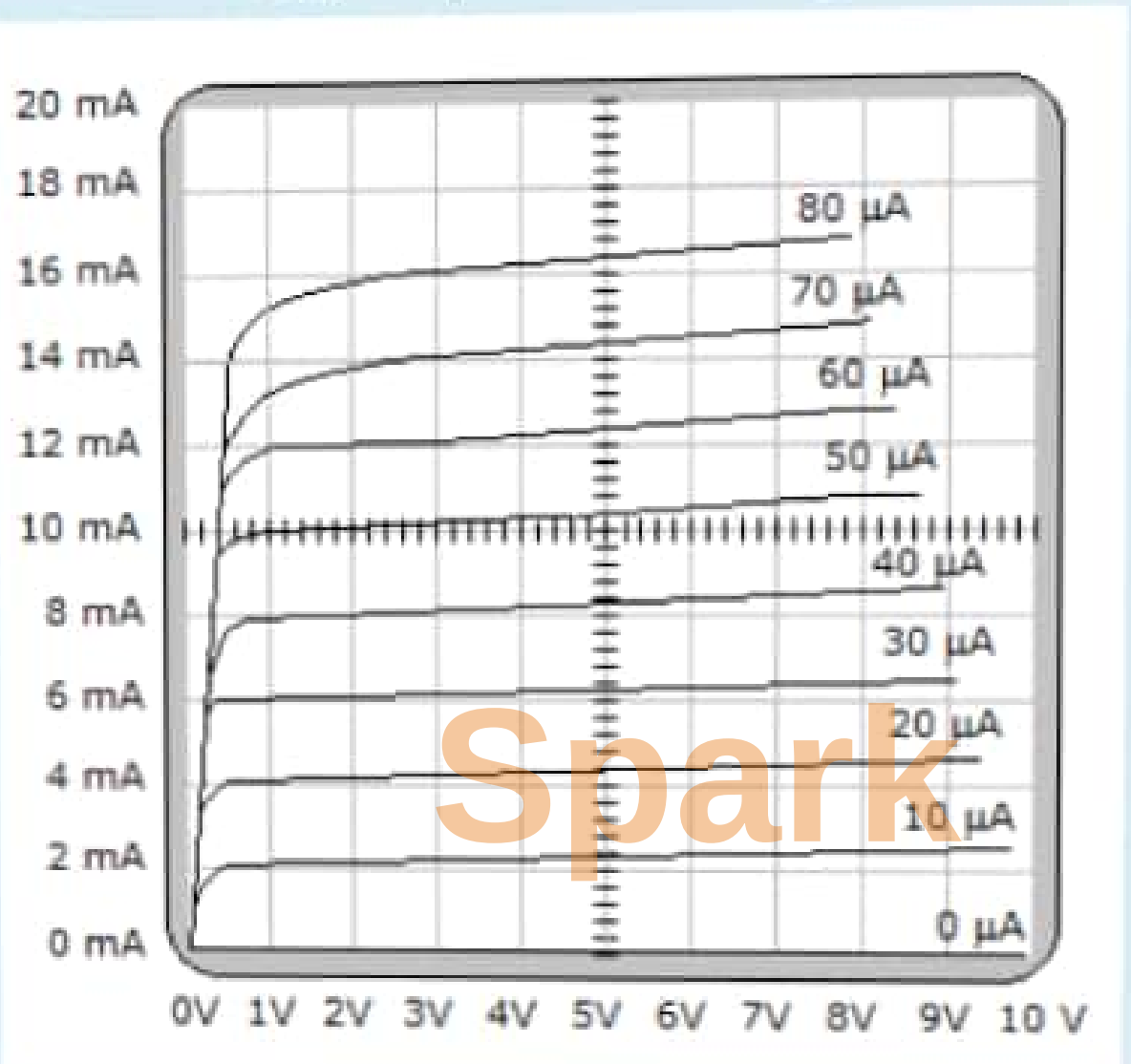
Select one:

- a. 4 times of input
- b. Equal to input
- c. Half of input
- d. Double of input

Spark

ة ضوئيا بـ CamScanner

Calculate β_{ac} for I_C = 15 mA and V_{CE} = 5 V.



Spark

Select one:

- a. 200
- b. 180
- c. 220
- d. None of the above

CamScanner ـة ضوئيا بـ

Number of DC sources needed to bias the OP- Amp is

Select one

- a. 2
- b. 4
- c. Self biased
- d. 1

Spark

ة ضوئيا بـ CamScanner

Series Limiter Characteristics

Select one:

- a. the output signal is in phase with the input signal
- b. The diode is connected in series with the signal path, polarized so that either the (-) or (+) portion of the signal will pass unaffected
- c. An output is taken across the resistor, connected across the output side of the diode
- d. all of the mentioned

Spark

ة ضوئيا بـ CamScanner

For V_{GS} = 0V, the drain current becomes constant when V_{DS} exceeds :

Select one:

- a. V_{DD}
- b. V_p
- c. 0V
- d. Cutoff

Spark

CamScanner ـة ضوئيا بـ

The gain of the inverting amplifier is obtained

Select one:

- a. Av = - Rf/Ri
- b. Av = - (1 + Rf/Ri)
- c. Av = 1 + Rf/Ri
- d. Av = Rf / Ri

Spark

ة ضوئيا بـ CamScanner

Is a type of clipper which removes the positive half cycles of the input AC signal.

Select one:

- a. shunt negative clipper
- b. dual clipper
- c. series negative clipper
- d. series positive clipper

Spark

ة ضوئيا بـ CamScanner

N - JFET Terminals when tested , results are :

Select one:

- a. Zero between Gate and Source
- b. Zero between Gate and Drain
- c. Zero (short) between Drain and Source
- d. 0.7 V between Drain and Source

Spark

ة ضوئيا بـ CamScanner

The Terminals not used in the OP- amp, they are

Select one:

- a. Terminals 4 , 7 , 6
- b. Terminals 1 , 5 , 8
- c. Terminals 2 , 3 , 4
- d. Terminals 1, 3, 5

Spark

ة ضوئيا بـ CamScanner

A filtered full-wave rectifier voltage has a smaller ripple than does a half-wave rectifier voltage for the same load resistance and capacitor values. This statement is

Select one:

- a. True
- b. False

Spark

ة ضوئيا بـ CamScanner

A no load conditions means that

Select one:

- a. none of these
- b. the load has zero resistance
- c. the load has infinite resistance
- d. the output terminals are grounded

Spark

ة ضوئيا بـ CamScanner

To get a peak load voltage of 40 Vout of a bridge amplifier. What is the approximate rms value of secondary voltage?

Select one:

- a. 56.6 V
- b. 14.4 V
- c. 28.3 V
- d. 0 V

Spark

ة ضوئيا بـ CamScanner

Which of the following regions is (are) part of the output characteristics of a transistor?

Select one:

- a. active
- b. cutoff
- c. all of the above
- d. saturation

Spark

ة ضوئيا بـ CamScanner

Is a type of clipper which removes some portion of both positive and negative half cycles of the input AC signal.

Select one:

- a. series negative clipper
- b. daul clipper
- c. shunt positive clipper
- d. series positive clipper

Spark

CamScanner ـة ضوئيا بـ

Reverse current will drastically increase when externalreverse biased voltage is increased to a Value called

Select one:

- a. break over voltage
- b. breakdown voltage
- c. drift voltage
- d. dynamic voltage

Spark

ة ضوئيا بـ CamScanner

Depending on the type of the applied voltage, its resistance at any point on the curve can be determined

Select one:

- a. drain resistance
- b. dynamic resistance
- c. base resistance
- d. static resistance

Spark

ة ضوئيا بـ CamScanner

For a properly biased *pnp* transistor, let I_C = 10 mA and I_E = 10.2 mA. What is the level of I_B?

Select one:

- a. 0.2 A
- b. 200 μA
- c. 200 mA
- d. 20.2 mA

Spark

CamScanner ـة ضوئيا بـ

For an OP- amp. with negative feedback, the output is

Select one:

- a. Fed back to the inverting input
- b. Equal to the input
- c. Increased
- d. Fed back to the noninverting input

Spark

ة ضوئيا بـ CamScanner

The constant current region of a FET lies between :

Select one:

- a. Cutoff and pinch - off
- b. 0 and IDSS
- c. Pinch - off and breakdown
- d. Cutoff and saturation

Spark

ة ضوئيا بـ CamScanner

The gain Av in C.D amplifier is:

Select one:

a. 1

b. ∞

c. Zero

d. None of all

Spark

ة ضوئيا بـ CamScanner

N - JFET Terminals when tested , results are :

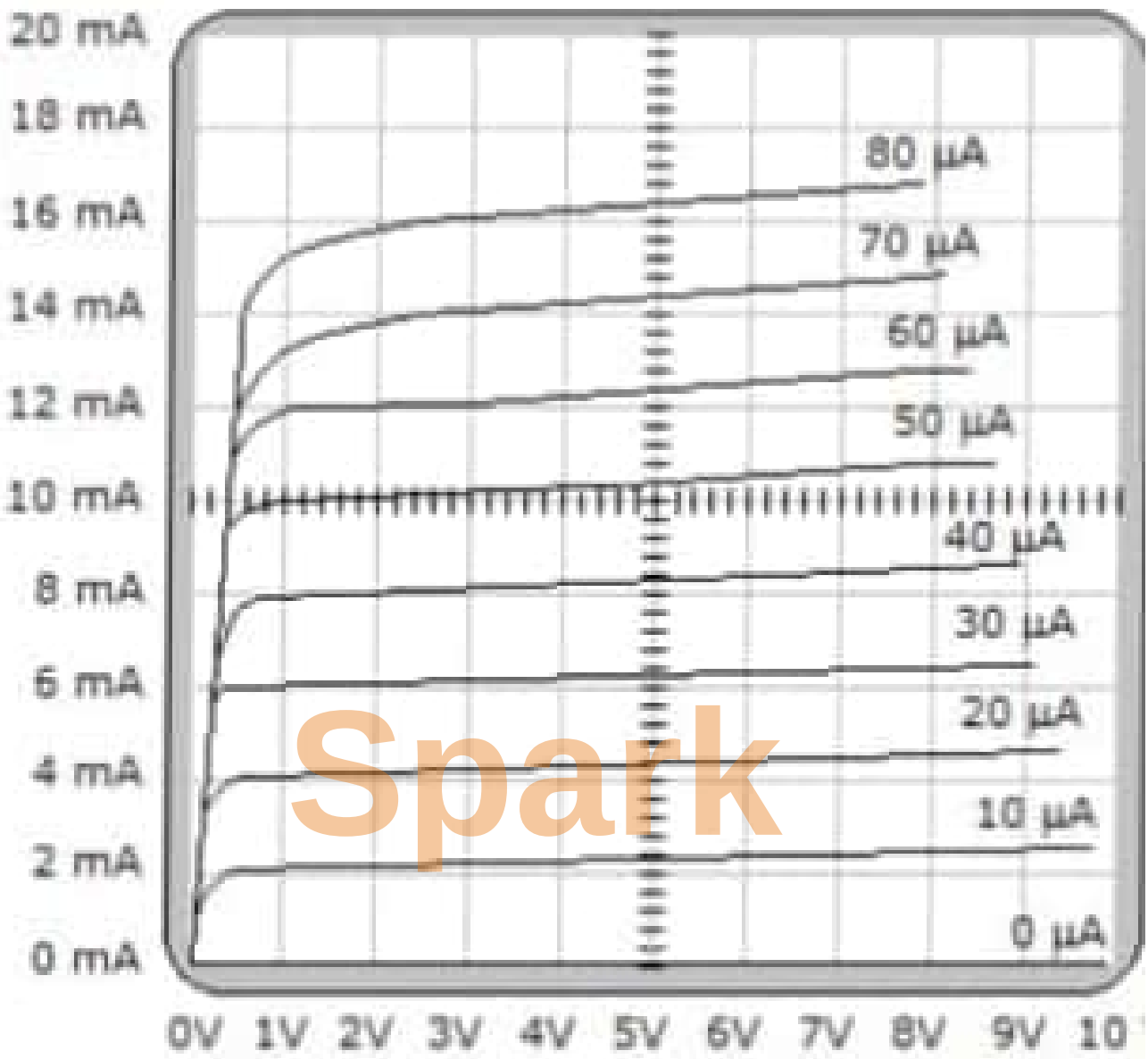
Select one:

- a. 0.7 V between Drain and Source
- b. Zero (short) between Drain and Source
- c. Zero between Gate and Source
- d. Zero between Gate and Drain

Spark

ة ضوئيا بـ CamScanner

Calculate β_{ac} for $I_C = 15$ mA and $V_{CE} = 5$ V.



Select one:

- a. 200
- b. 180
- c. 220
- d. None of the above

CamScanner بـ ضوئيا ة

To get a peak load voltage of 40 Vout of a bridge amplifier. What is the approximate rms value of secondary voltage?

Select one:

- a. 28.3 V
- b. 14.4 V
- c. 0 V
- d. 56.6 V

Spark

CamScanner ـة ضوئيا بـ

The integrated OP-Amp. has

Select one:

- a. One input and one output
- b. Two inputs and two outputs
- c. One input and two outputs
- d. Two inputs and one output

Spark

ة ضوئيا بـ CamScanner

When is a regulator used?

Select one:

- a. when there are large variations in load current and input voltage
- b. when there are small variations in load current and input voltage
- c. when there are no variations in load current and input voltage
- d. when there are small variations in load current and large variations in input voltage

Spark

ة ضوئيا بـ CamScanner

A transistor can be checked using a(n) _____

Select one:

a. digital meter

b. Any of the above

c. curve tracer

d. ohmmeter

Spark

ة ضوئيا بـ CamScanner

A noninverting amplifier has Ri of 1KΩ and Rf of 100 KΩ. The closed loop gain is

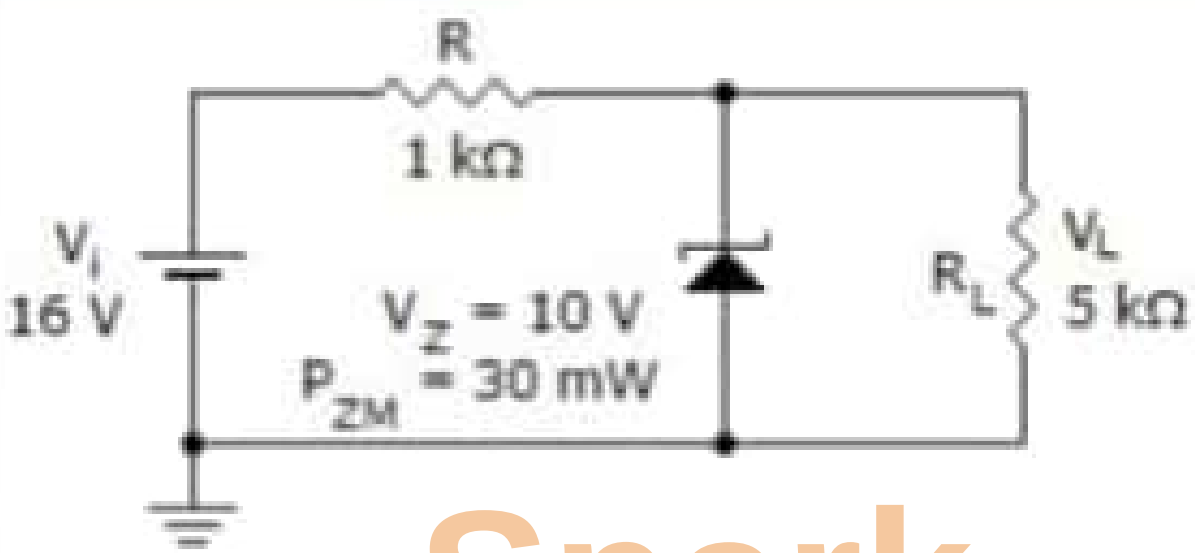
Select one:

- a. 100000
- b. 100
- c. 101
- d. 1000

Spark

ة ضوئيا بـ CamScanner

Calculate I_L and I_Z



Spark

Select one:

a. 2 mA, 4 mA

b. 2 mA, 0 mA

c. 4 mA, 2 mA

d. 4 mA, 2 mA

ة ضوئيا بـ CamScanner

Which of the following regions is (are) part of the output characteristics of a transistor?

Select one:

- a. active
- b. saturation
- c. cutoff
- d. all of the above

Spark

CamScanner ـة ضوئيا بـ

Term refers to the rise of DC voltage to establish certain operating conditions for an electronic

Select one:

- a. bias
- b. input
- c. source
- d. core

Spark

ة ضوئيا بـ CamScanner

To get a peak load voltage of 40 Vout of a bridge amplifier. What is the approximate rms value of secondary voltage?

Select one:

- a. 14.4 V
- b. 0 V
- c. 56.6 V
- d. 28.3 V

Spark

ة ضوئيا بـ CamScanner

A filtered full-wave rectifier voltage has a smaller ripple than does a half-wave rectifier voltage for the same load resistance and capacitor values. This statement is

Spark

Select one:

- a. True
- b. False

ة ضوئيا بـ CamScanner

Series Limiter Characteristics

Select one:

- a. the output signal is in phase with the input signal.
- b. The diode is connected in series with the signal path, polarized so that either the (-) or (+) portion of the signal will pass unaffected.
- c. An output is taken across the resistor, connected across the output side of the diode.
- d. all of the mentioned

Spark

CamScanner ـة ضوئيا بـ

The diode characteristic curve is a plot between

Select one:

- a. voltage and temprature
- b. voltage and time
- c. current and time
- d. voltage and current

Spark

ة ضوئيا بـ CamScanner

If a 169.7 V half-wave peak has an average voltage of 54 V, what is the average of two full-wave peaks?

Select one:

- a. 339.4 V
- b. 115.7 V
- c. 108.0 V
- d. 119.9 V

Spark

ة ضوئيا بـ CamScanner

To get a peak load voltage of 40 Vout of a bridge amplifier. What is the approximate rms value of secondary voltage?

Select one:

- a. 28.3 V
- b. 14.4 V
- c. 56.6 V
- d. 0 V

Spark

ة ضوئيا بـ CamScanner

Is a type of clipper which removes some portion of both positive and negative half cycles of the input

Select one:

- a. series positive clipper
- b. shunt positive clipper
- c. series negative clipper
- d. daul clipper

Spark

ة ضوئيا بـ CamScanner

To get a peak load voltage of 40 Vout of a bridge amplifier. What is the approximate rms value of secondary voltage?

Select one:

- a. 28.3 V
- b. 14.4 V
- c. 0 V
- d. 56.6 V

Spark

ة ضوئيا بـ CamScanner

Diode rectifier is a circuit that causes an ac input voltage to be converted into a pulsed waveform having an average, or dc output voltage. This statement is

Select one:

- a. False
- b. True

Spark

ة ضوئيا بـ CamScanner

How many diode in Center-tap Full wave rectifier

Select one:

- a. 2
- b. 2 or 4
- c. 4
- d. 1

Spark

ة ضوئيا بـ CamScanner

In a half wave rectifier, the sine wave input is 200sin300t. The average value of output voltage is?

Select one:

- a. 76.987V
- b. 57.876V
- c. 63.694V
- d. 67.453V

Spark

ة ضوئيا بـ CamScanner

In forward bias, positive terminal of source is connected to diode's positive terminal referred to as

Select one:

- a. cathode
- b. supply
- c. input
- d. anode

Spark

ة ضوئيا بـ CamScanner

If a 169.7 V half-wave peak has an average voltage of 54 V, what is the average of two full-wave peaks?

Select one:

- a. 339.4 V
- b. 115.7 V
- c. 108.0 V
- d. 119.9 V

Spark

CamScanner ـة ضوئيا بـ

A properly functioning diode in forward biased will produce a reading of

Select one:

- a. 0.2-0.7 V
- b. 0.5-5 V
- c. 0.5-0.9 V
- d. 0.5-9 V

Spark

ة ضوئيا بـ CamScanner

If biasing voltage is 10 V and R=1.0 kΩ than forward voltage for practical diode model will be

Select one:

- a. 10.7 V
- b. 5 V
- c. 10 V
- d. 9.3 V

Spark

CamScanner ـة ضوئيا بـ

How many diode in Center-tap Full wave rectifier

Select one:

- a. 2
- b. 1
- c. 2 or 4
- d. 4

Spark

ة ضوئيا بـ CamScanner

Reverse current will drastically increase when externalreverse biased voltage is increased to a Value called

Select one:

- a. break over voltage
- b. dynamic voltage
- c. breakdown voltage
- d. drift voltage

Spark

ة ضوئيا بـ CamScanner

A clipper is a device that removes some portion of the input DC signal. This statement is

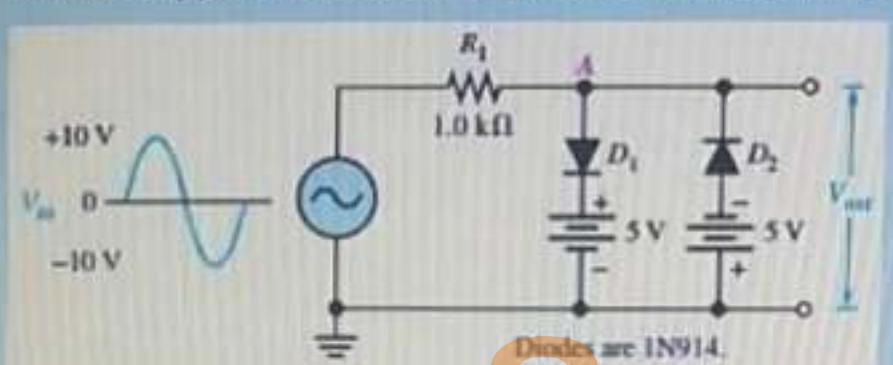
Select one:

- a. True
- b. False

Spark

ة ضوئيا بـ CamScanner

For the clipper circuit below, the output will be: (in order)



Spark

Select one:

- a. -5.7,5.7
- b. 5.7,5.7
- c. 5.7,-5.7
- d. -5.7,-5.7

CamScanner ـة ضوئيا بـ

A properly functioning diode in forward biased will produce a reading of

Select one:

- a. 0.5-0.9 V
- b. 0.5-9 V
- c. 0.2-0.7 V
- d. 0.5-5 V

Spark

ة ضوئيا بـ CamScanner

If biasing voltage is 10 V and R=1.0 kΩ than forward voltage for practical diode model will be

Select one

- a. 9.3 V
- b. 5 V
- c. 10 V
- d. 10.7 V

Spark

ة ضوئيا بـ CamScanner

In an inverting op-amp circuit for which the gain is −4 V/V and the total resistance used is 100 kΩ. Then the value of R_1 and R_2 (negative feedback)

Select one:

- a. R_1 = 80KΩ and R_2 = 20KΩ
- b. R_1 = 20KΩ and R_1 = 80KΩ
- c. R_1 = 40KΩ and R_1 = 60KΩ
- d. R_1 = 50KΩ and R_1 = 50KΩ

Spark

ة ضوئيا بـ CamScanner

Output Impedance of Common Drain Amplifier is_____

Select one:

- a. $R_S \| r_d$
- b. R_S
- c. $R_S + r_d$
- d. $R_S \| 1/g_m$

Spark

ة ضوئيا بـ CamScanner

The collector supply voltage for a CE configured transistor is 10V. The resistance R_L=800Ω. The voltage drop across R_L is 0.8V. Find the value of collector emitter voltage

Select one:

- a. 6.5V
- b. 3.7V
- c. 9.8V
- d. 9.2V

Spark

ة ضوئيا بـ CamScanner

Which of the following is true about Common Drain amplifier?

Select one:

- a. It has low input impedance
- b. Infinite gain
- c. Output will be same as input
- d. It has high output impedance

Spark

ة ضوئيا بـ CamScanner

For an ideal non-inverting operational amplifier having finite gain (A), the ratio of output voltage (v_0) to input voltage (v_i) is (given R_2 is the feedback resistance)

Select one:

- a. (R2/R1)/(((1+R2/R1)/A))
- b. (1+R2/R1)/(1+((1+R2/R1)/A))
- c. (R2/R1)/(1+((1+R2/R1)/A))
- d. (1+R2/R1)/(((1+R2/R1)/A))

Spark

ة ضوئيا بـ CamScanner

. For an ideal negative feedback configuration which of the following is true?

Select one:

- a. The closed loop gain for a negative feedback does not depend only on the external parameters
- b. There is a virtual ground at the negative input terminal
- c. There is a virtual open circuit between the input terminals
- d. There is a virtual short circuit between the input terminals

Spark

ة ضوئيا بـ CamScanner

. For ideal non-inverting operational amplifier

Select one:

- a. Input resistance is zero and output resistance is infinite
- b. Input and output resistances are zero
- c. Input and output resistances are infinite
- d. Input resistance is infinite and output resistance is zero

Spark

CamScanner ـة ضوئيا بـ

n CE configuration, if the voltage drop across 5kΩ resistor connected in the collector circuit is 5V. Find the value of I_B when β=50.

Select one:

- a. 0.03mA
- b. 0.01mA
- c. 0.02mA
- d. 0.25mA

Spark

ة ضوئيا بـ CamScanner

The effect of the inverting configuration is

Select one:

- a. The output signal and the input signal are in phase
- b. The output signal and the input signal are out of phase by 180°
- c. The output phase is leading the input phase by 90°
- d. The output phase is lagging behind the input phase by 90°

Spark

ة ضوئيا بـ CamScanner

A transistor is connected in CE configuration. Collector supply voltage V_{cc}=10V, R_L=800Ω, voltage drop across R_L=0.8V, α=0.96. What is base current?

Select one:

- a. 56.78μA
- b. 67.67μA
- c. 41.97μA
- d. 78.54μA

Spark

ة ضوئيا بـ CamScanner

Which of the following is true about Common Drain amplifier?

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Spark

ة ضوئيا بـ CamScanner

. For ideal non-inverting operational amplifier

Select one:

- a. Input resistance is zero and output resistance is infinite
- b. Input and output resistances are zero
- c. Input and output resistances are infinite
- d. Input resistance is infinite and output resistance is zero

Spark

CamScanner ـة ضوئيا بـ

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Spark

ة ضوئيا بـ CamScanner

Refer to approximate output waveform waveform in fig. 1, select the correct circuit arrangement in fig.2 that will give this approximate output waveform?

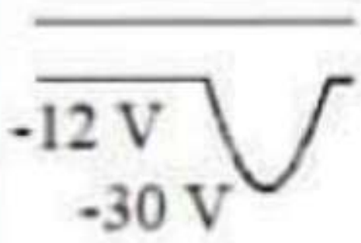
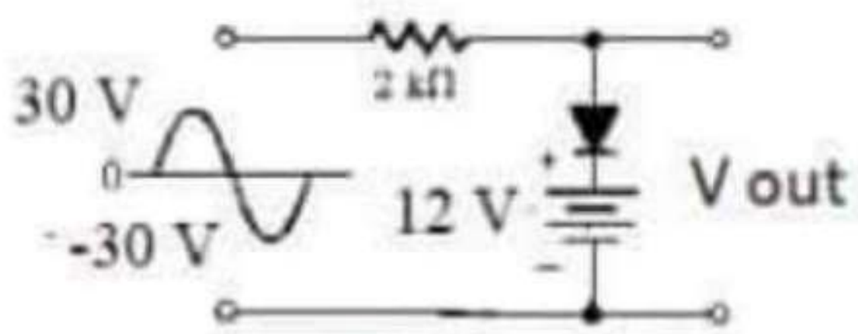
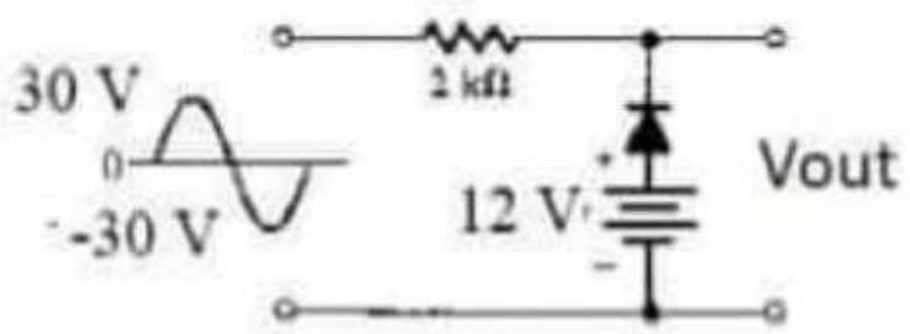


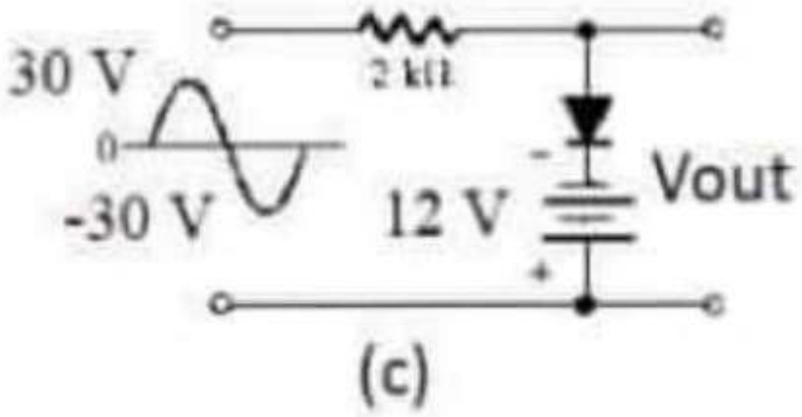
Fig. 1



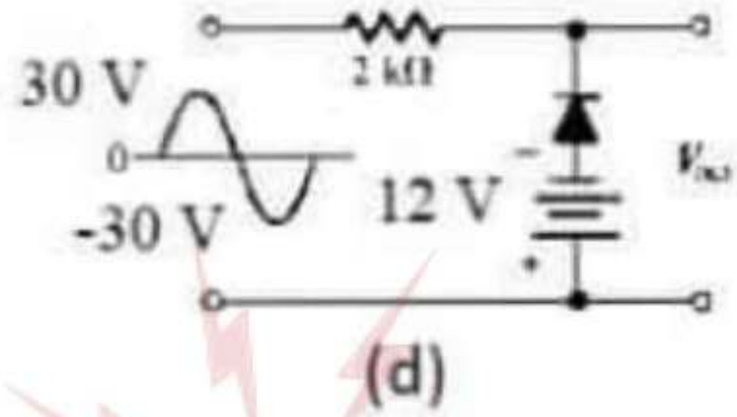
(a)



(b)



(c)



(d)

Fig. 2

Select one:

- a. a
- b. b
- c. c
- d. d

الممسوحة ضوئيا بـ CamScanner

Refer to approximate output waveform waveform in fig. 1, select the correct circuit arrangement in fig.2 that will give this approximate output waveform?

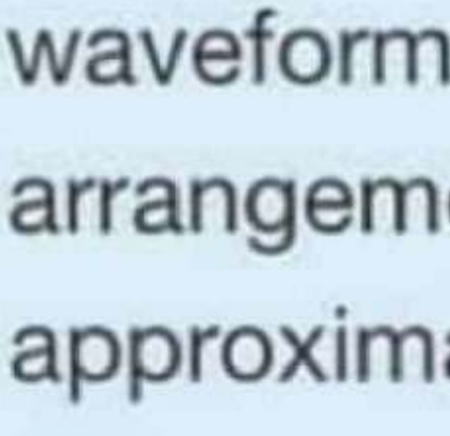


Fig. 1

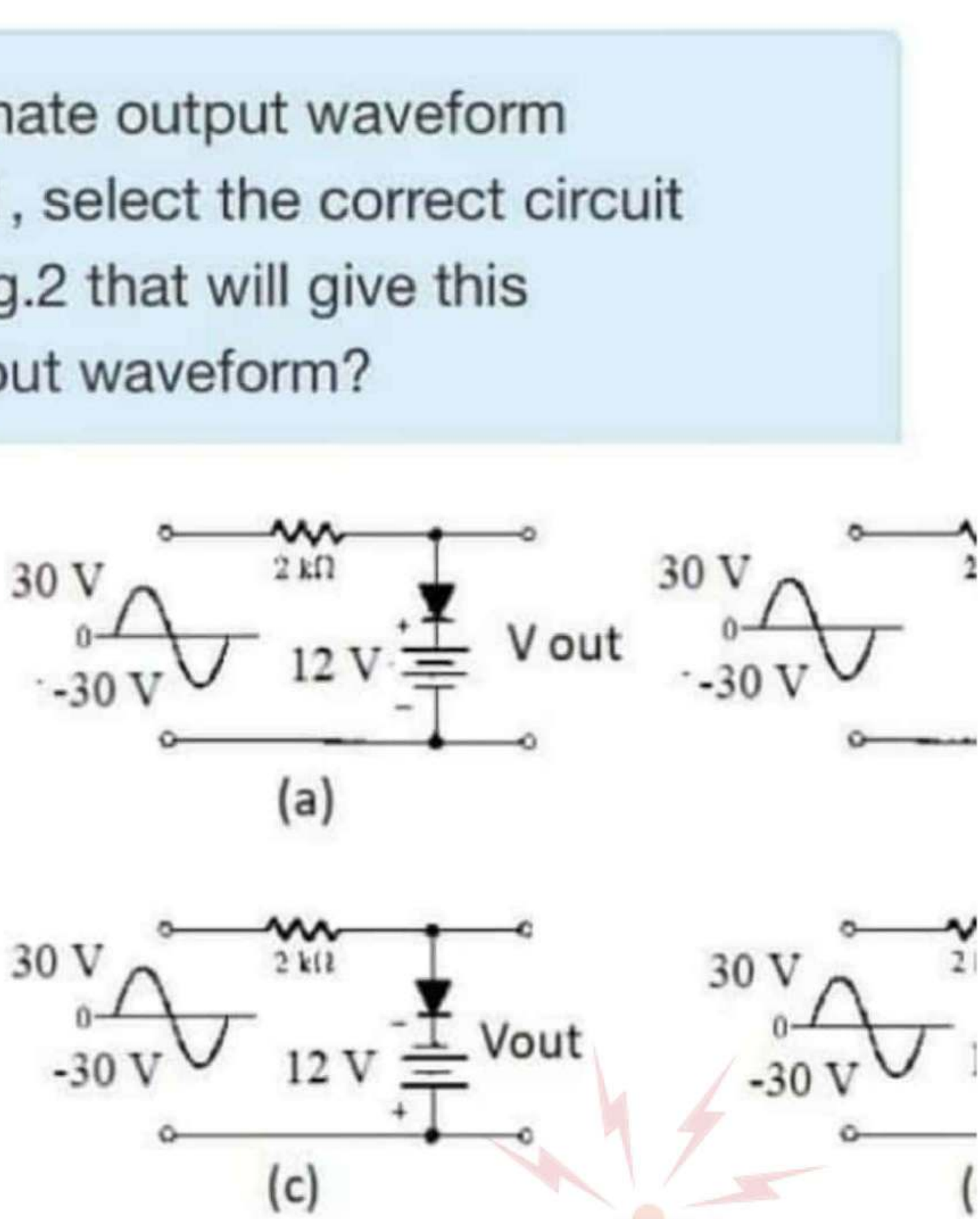


Fig. 2

Select one:

- a. a
- b. b
- c. c
- d. d

POLYTECHNIC

الممسوحة ضوئيا بـ CamScanner

Refer to approximate output waveform waveform in fig. 1, select the correct circuit arrangement in fig.2 that will give this approximate output waveform?

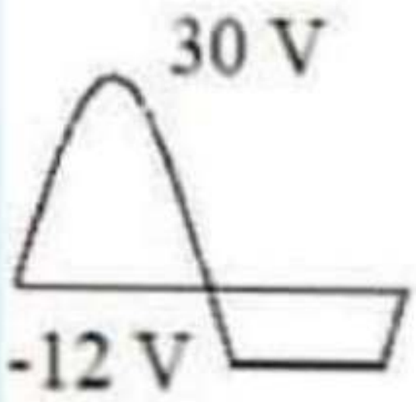


Fig. 1

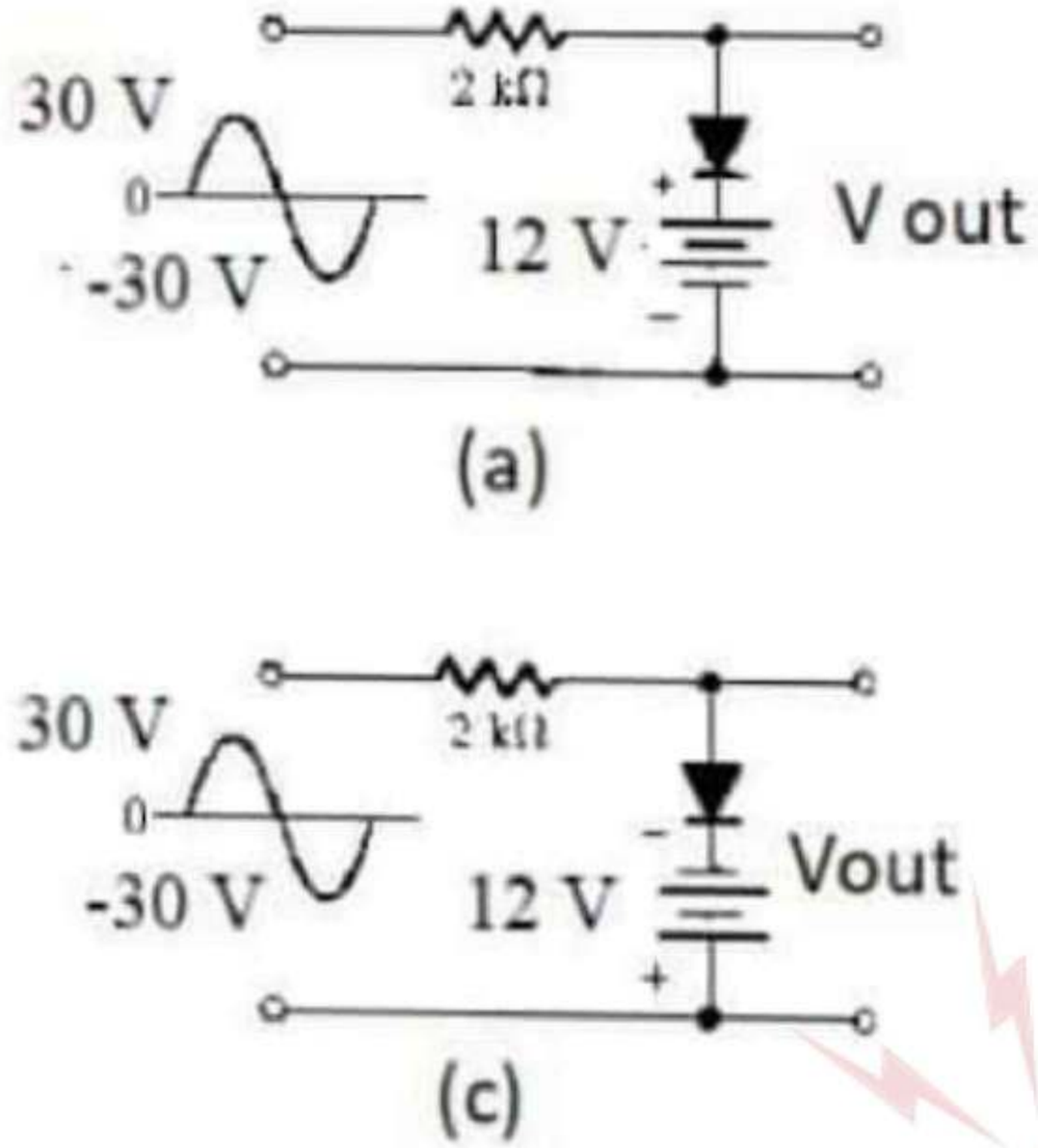


Fig. 2

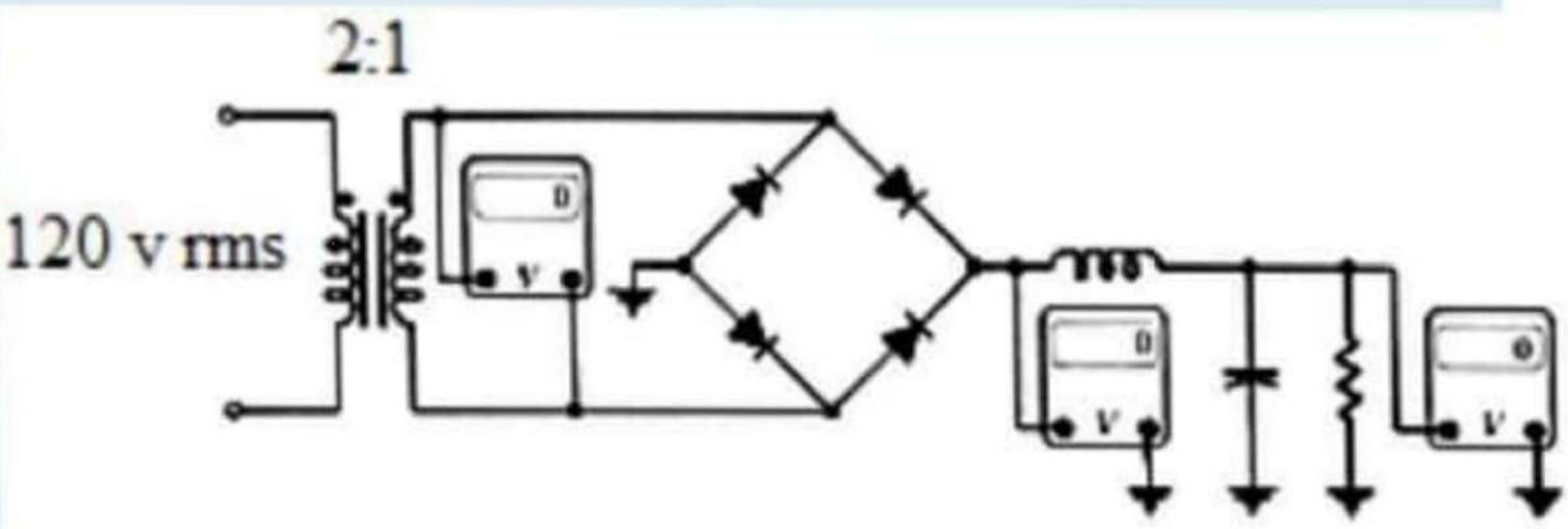
Select one:

- a. a
- b. b
- c. c
- d. d

POLYTECHNIC

الممسوحة ضوئيا بـ CamScanner

Refer to Figure shown. If the voltmeter across the transformer reads 0 V, the probable trouble, if any, would be



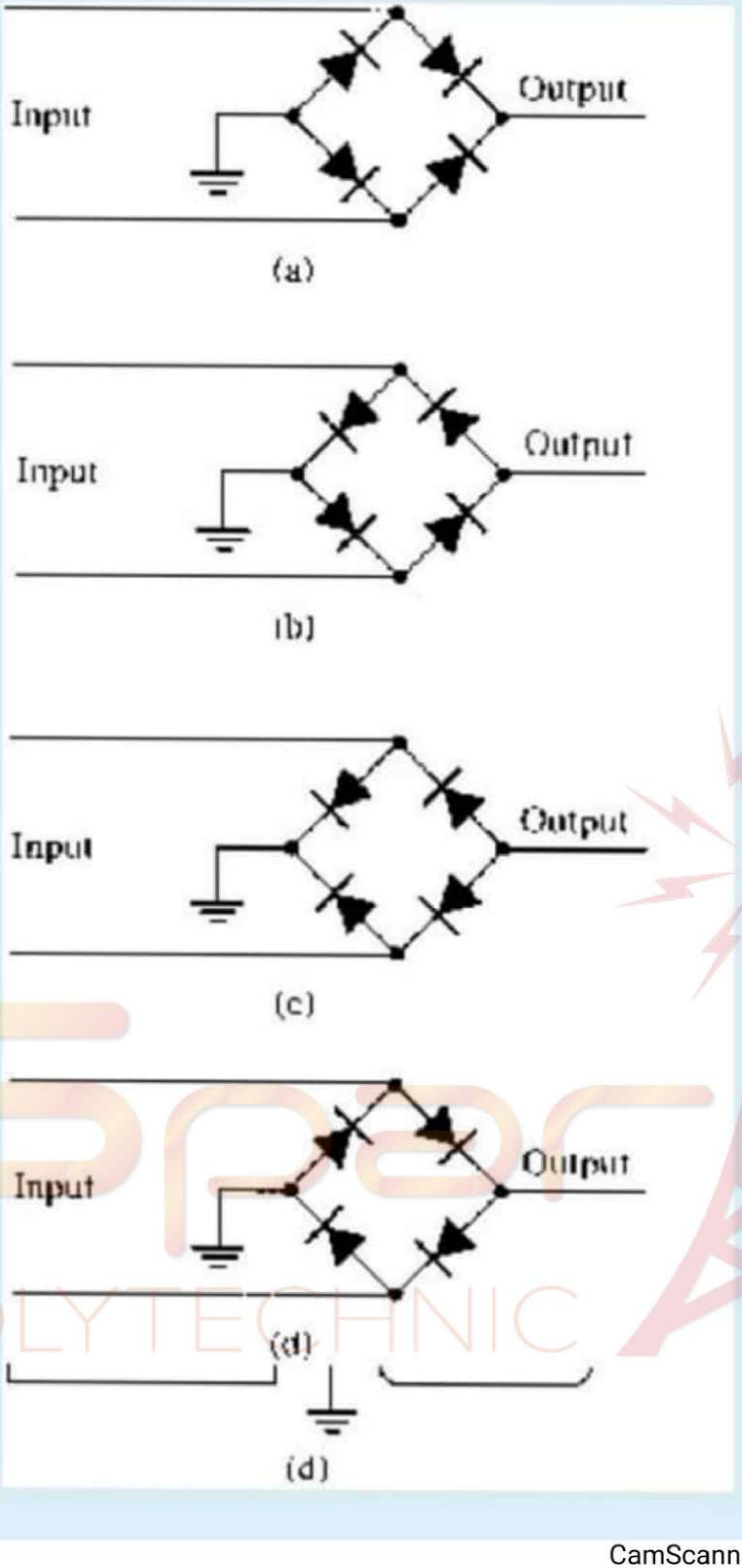
Select one:

- a. an open transformer secondary.
- b. one of the diodes is open.
- c. the filter capacitor is shorted.
- d. no trouble exists.
- e. a diode is shorted.

الممسوحة ضوئيا بـ CamScanner

الممسوحة ضوئيا بـ CamScanner

Refer to Figure shown. Which diode arrangement will supply a positive output voltage?



الممسوحة ضوئيا بـ CamScanner

In the operation of a half-wave rectifier with a capacitor-input filter, the ripple factor can be lowered by _____ the value of the filter capacitor or _____ the load resistors.

Select one:

- a. decreasing, increasing
- b. decreasing, decreasing
- c. increasing, increasing
- d. increasing, decreasing

الممسوحة ضوئيا بـ CamScanner

A diode conducts current when reverse-biased and blocks current when

forward-biased

Select one:

- a. True
- b. False

POLYTECHNIC

الممسوحة ضوئيا بـ CamScanner

The diode characteristic curve is a plot between

Select one:

- a. current and time
- b. voltage and temprature
- c. voltage and time
- d. voltage and current

POLYTECHNIC

الممسوحة ضوئيا بـ CamScanner

CamScanner الممسوحة ضوئيا بـ

Actual resistance of a forward biased diode is typically much less than

Select one:

- a. 100 Ω
- b. 500 Ω
- c. 100k Ω
- d. 500 kΩ

POLYTECHNIC

الممسوحة ضوئيا بـ CamScanner

الممسوحة ضوئيا بـ CamScanner

Condition in which diode never intentionally operate is

Select one:

- a. reverse biased
- b. neutral biased
- c. forward biased
- d. floating biased

الممسوحة ضوئيا بـ CamScanner

ıpproximate output waveform
ı in fig. 1, select the correct circuit
ent in fig.2 that will give this
ate output waveform?

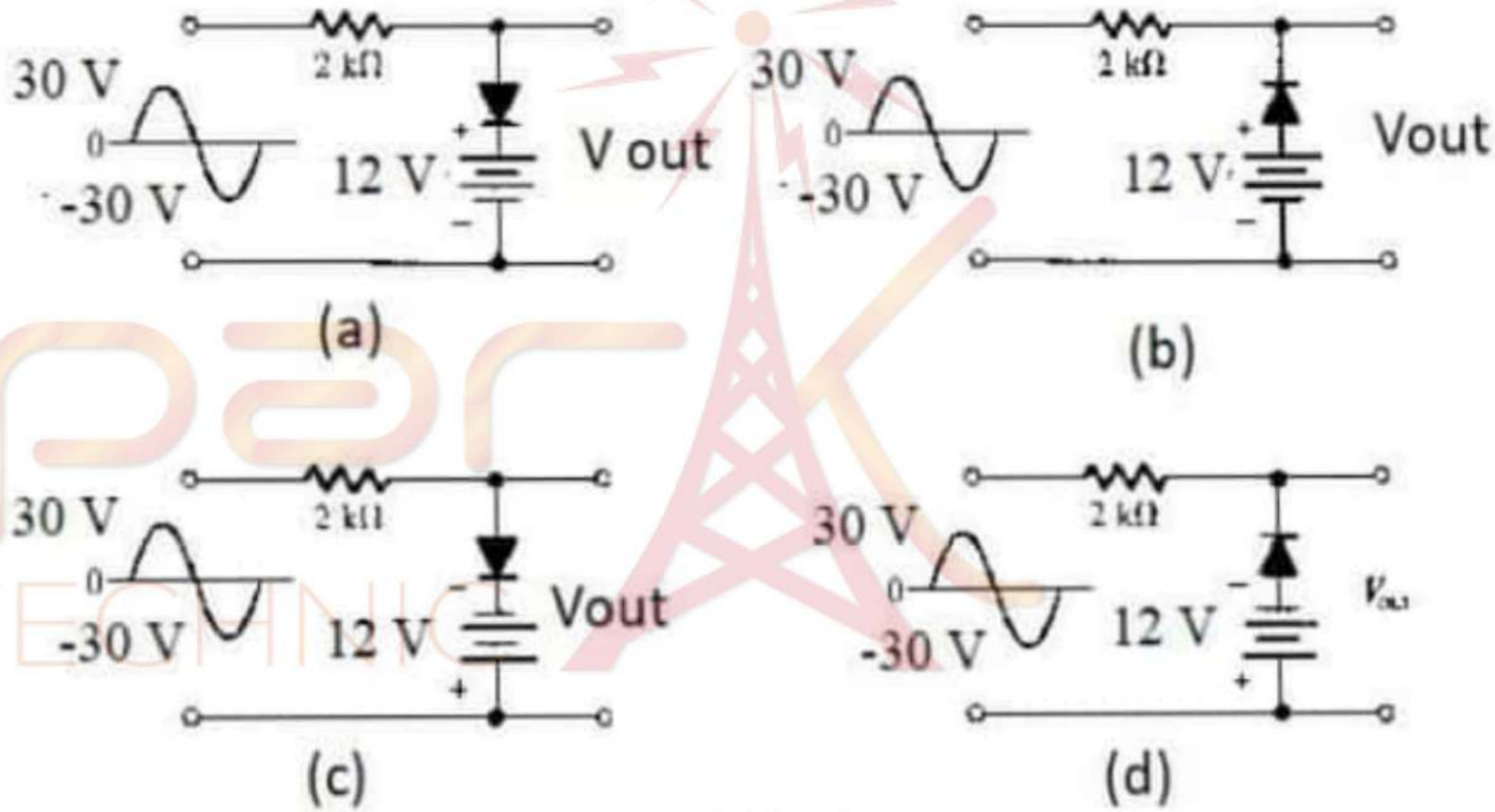


Fig. 2

POLYTECHNIC
الممسوحة ضوئيا بـ CamScanner
CamScanner بـ ضوئيا ة

When transistors are used in digital circuits they usually operate in the:

Select one:

- a. saturation and cutoff regions
- b. linear region
- c. active region
- d. breakdown region

If a 169.7 V half-wave peak has an average voltage of 54 V, what is the average of two full-wave peaks?

Select one:

- a. 119.9 V
- b. 108.0 V
- c. 115.7 V
- d. 339.4 V

Spark POLYTECHNIC

Is a type of clipper which removes the negative half cycles of the input AC signal.

Select one:

- a. series negative clipper
- b. series positive clipper
- c. shunt positive clipper
- d. daul clipper

Spark POLYTECHNIC

For a clamping circuit , Sometimes an independent dc supply is also required to cause an:

Select one:

- a. additional V peak output ,
- b. additional shift.
- c. additional frequency.
- d. additional V peak input .

Spark POLYTECHNIC

When the regulation by a Zener diode is with a varying input voltage, what happens to the voltage drop across the resistance?

Select one:

- a. Increases
- b. Has no effect on voltage
- c. The variations depend on temperature
- d. Decreases

Spark POLYTECHNIC

The most frequent application for a _____ is in regulator networks and as a reference voltage.

Select one:

- a. Zener diode
- b. half-wave rectifier
- c. full-wave rectifier
- d. ideal diode

Spark POLYTECHNIC

In a certain biased limiter, the bias voltage is 5 V, and the input is 10 V peak sine wave. If the positive terminal of the bias voltage is connected to the cathode of the diode, the maximum voltage ot the anode is:

Select one:

- a. 10 V
- b. 5.7 V
- c. 5 V
- d. 0.7 V

Spark POLYTECHNIC

Is a type of clipper which removes the positive half cycles of the input AC signal.

Select one:

- a. shunt negative clipper
- b. series negative clipper
- c. series positive clipper
- d. dual clipper

Diode type that is used to regulate the output voltage is

Select one:

- a. Silicon Diode
- b. Zener
- c. lED
- d. Germanium

Spark POLYTECHNIC

To measure a current passes through an element, the Ammeter is placed in:

Select one:

- a. parallel
- b. No effect
- c. series
- d. (B and C)

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A filtered full-wave rectifier voltage has a smaller ripple than does a half-wave rectifier voltage for the same load resistance and capacitor values. This statement is

Select one:

- a. False
- b. True

Diode clamper is:

Select one:

- a. Change the voltage level of an dc signal by adding a ac voltage
- b. Change the voltage level of an ac signal by adding a ac voltage
- c. Change the voltage level of an dc signal by adding a dc voltage
- d. Change the voltage level of an ac signal by adding a dc voltage

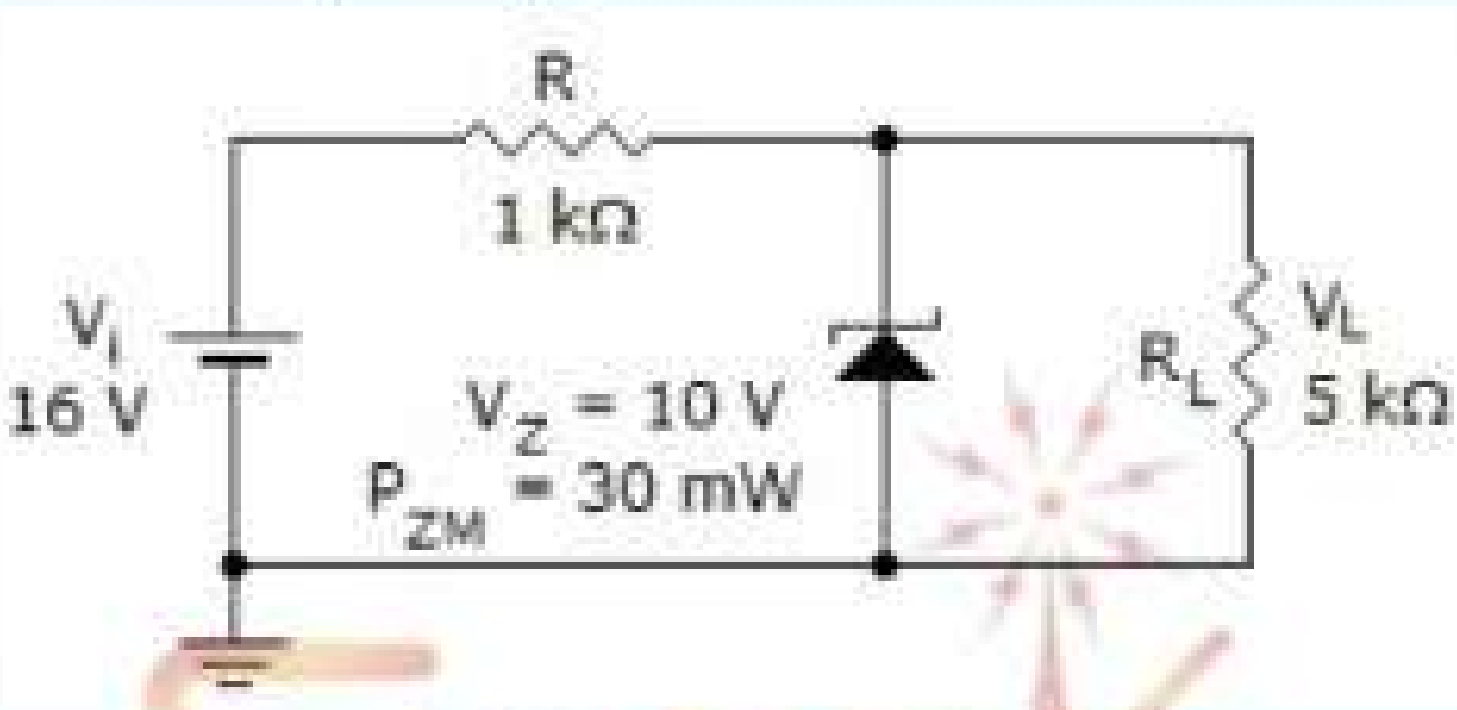
To get a peak load voltage of 40 Vout of a bridge amplifier. What is the approximate rms value of secondary voltage?

Select one:

- a. 0 V
- b. 14.4 V
- c. 28.3 V
- d. 56.6 V

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Calculate I_L and I_Z



Select one:

- a. 2 mA, 0 mA
- b. 4 mA, 2 mA
- c. 4 mA, 2 mA
- d. 2 mA, 4 mA

A zener diode can be considered as : .

Select one:

- a. Is forward biased
- b. Has a barrier potential of 1v
- c. A battery
- d. Has a constant voltage in the breakdown region

Depending on the type of the applied voltage, its resistance at any point on the curve can be determined

Select one:

- a. base resistance
- b. drain resistance
- c. static resistance
- d. dynamic resistance

clamper circuit is:

Select one:

- a. Usually Capacitance and resistance combination at the input
- b. Usually diode and resistance parallel combination at the output
- c. Usually Capacitance and resistance combination in parallel
- d. Usually Capacitance and diode parallel combination at the input

When in its "on" state, the voltage across an ideal Zener diode, _____.

Select one:

- a.
  gets larger with an increase in applied voltage
- b. increases sharply with a decrease in applied voltage
- c.
  gets smaller with an increase in applied voltage
- d.
  None of these

When measuring an input Ac voltage using the DMM, the reading is given :

Select one:

- a. Peak Value
- b. Peak to Peak value
- c. RMS value
- d. All of Above